The De-dollarization Movement: How BRICS Reserve Changes Impact US Dollar Investments and Portfolio Risk

Executive Summary

This report examines BRICS-related reserve changes, their implications for U.S. Treasury yields and bond prices, and the diversification role of gold in a stock–bond portfolio.

De-dollarization is observable, but it is neither uniform across BRICS members nor equivalent to wholesale liquidation of U.S. assets. Between December 2018 and December 2025, mainland China's reported Treasury position fell from US$1.124 trillion to US$684.4 billion and Brazil's from US$303.1 billion to US$168.7 billion. India's reported holdings, by contrast, rose from US$141.3 billion to US$182.9 billion. Over the same period, total foreign Treasury holdings increased from US$6.270 trillion to US$9.270 trillion (US$9,270.477 billion before rounding). The reported positions of Russia and South Africa are much smaller, with Russia's initial sharp reduction occurring in April–May 2018. These facts reject any simple proposition that "BRICS" behaves as one homogeneous reserve manager (U.S. Department of the Treasury, 2026c).

The second important result concerns **measurement**. A fall in a country's reported Treasury holdings is not necessarily equal to a transaction flow. TIC records can be affected by securities valuation, repayment, transactions and the location of foreign custodians. The U.S. Treasury itself cautions that securities held through custodians outside the investor's country may not be attributed to the ultimate beneficial owner. The Federal Reserve Bank of Kansas City further shows that, for aggregate foreign official investors, much of the decline in the market value of Treasury holdings since 2020 reflected valuation rather than active net selling. Therefore, a causal statement such as "BRICS sales raised the 10-year yield by X basis points" cannot be inferred from country holdings alone (Dilts Stedman, 2025).

The country-level CSLT decomposition attributes US$533.6 billion of China's US$555.8 billion long-term Treasury decline over calendar 2018–2025 to net sales, with valuation and other changes reconciling the difference. The country-level and aggregate official-investor results can differ because the country series include private holders, use different periods, and contain estimated transactions through January 2023 (McCallum et al., 2026).

A similar distinction applies to foreign-exchange reserves. IMF COFER is a **global aggregate dataset**, not a BRICS-country dataset, and it excludes monetary gold. The September 2026 release puts the U.S. dollar at 56.52% of global foreign-exchange reserves in 2025 Q4, down from 62.07% in 2018 Q4; the renminbi remained near 1.96%. In a separately preserved October 2025 publication snapshot, the IMF showed that 92% of the apparent 1.47-percentage-point decline in the dollar share during 2025 Q2 was caused by exchange-rate valuation effects; with exchange rates held constant, the decline would have been only 0.12 percentage points. Thus, changes in reported reserve shares must be decomposed before being interpreted as deliberate de-dollarization (International Monetary Fund, 2026a; Kwende et al., 2025).

The evidence nevertheless supports a broader diversification interpretation. Central-bank gold demand remained historically high: World Gold Council estimates put 2025 net purchases at 863 tonnes, following three consecutive years above 1,000 tonnes. Survey evidence also shows strong reserve-manager interest in gold, although survey expectations are not equivalent to completed purchases (World Gold Council, 2025, 2026a).

For U.S. Treasury yields, the strongest defensible conclusion is **conditional rather than causal**. Published empirical studies consistently find that foreign official demand can affect Treasury yields. Warnock and Warnock (2009) estimated that the 10-year yield would have been roughly 80 basis points higher without foreign-government accumulation of U.S. government bonds over the twelve months ending May 2005, while later research produces different magnitudes depending on identification, maturity and whether flows or holdings are used. A 2025 Kansas City Fed synthesis translates a one-standard-deviation monthly foreign Treasury liquidation—approximately US$141 billion—into an average estimated yield impact of 57 basis points, with study estimates spanning approximately 25–101 basis points. These estimates are **not BRICS-specific coefficients**, so using 57 basis points as "the BRICS effect" would be methodologically wrong (Dilts Stedman, 2025; Warnock & Warnock, 2009).

At the investment level, duration provides a transparent bridge from macroeconomic scenarios to portfolio risk. A bond with modified duration of 7.5 years would lose approximately 1.88%, 3.75%, 5.63% and 7.50% under instantaneous parallel yield increases of 25, 50, 75 and 100 basis points respectively, before convexity, coupon accrual and reinvestment effects. These are **modeled scenarios**, not forecasts of losses caused by BRICS.

Finally, gold can improve portfolio efficiency, but the claim is conditional on weights, sample period and the asset from which gold is funded. A State Street backtest using monthly S&P 500 Total Return, Bloomberg U.S. Aggregate Bond Total Return and GLD data from December 2004 through December 2025 found that an even-funded 10% gold allocation raised the Sharpe ratio from 0.64 to 0.73 and reduced annualized volatility from 9.34% to 8.90%; when gold was funded entirely from bonds, the Sharpe ratio rose to 0.72 but volatility **increased** to 9.53%. The results show that funding choices affect portfolio volatility (Doshi et al., 2026).

The resulting investment conclusion is therefore not "sell dollars and buy gold." A more defensible conclusion is that **gradual reserve diversification creates an additional source of duration and regime risk for dollar portfolios, while a modest gold allocation can improve diversification in some environments; neither conclusion establishes that U.S. Treasury securities have ceased to function as core safe and liquid assets.**

The expanded RQ2 evidence distinguishes who absorbs additional Treasury supply and which maturity is affected. Official long-term net transactions were about **US$78.9 billion** over 2018–2025, while country-attributed flows mix official and private investors. Stablecoin-related bill demand and a Federal Reserve supply-shock model provide context, but neither establishes a BRICS-specific yield effect (Ahmed & Aldasoro, 2026; Beltran & Li, 2026).

For RQ3, an independently calculated 2018–2025 ETF exercise provides a second historical sample. Its 60/40 baseline returned 9.38% annually with 10.97% volatility; the 10% even-funded gold portfolio returned 10.26% with 10.34% volatility. The 10% bond-funded case had approximately unchanged volatility at 10.97%, whereas State Street's longer published sample showed volatility rising from 9.34% to 9.53%. Different periods, instruments and methods prevent interpreting this difference as a contradiction or a prediction (Doshi et al., 2026; Yahoo Finance, n.d.).

Data Analysis

Research period. The primary sample runs from **January 2018 to December 2025**. December 2017 provides the opening balance for calendar-2018 changes. The annual holdings tables compare December 2018 with December 2025, while monthly charts begin in January 2018. Gold and COFER comparisons cover Q4 2018 to Q4 2025. Russia's end-2017 currency composition is an opening baseline. Publications and observations from 2026 are identified as contextual.

Country and data-vintage scope. The study concerns the five founding BRICS members. Three-country annual Treasury charts cover China, India and Brazil; four-country monthly charts also cover Russia and exclude South Africa. Treasury calculations use TIC/CSLT data, including a December 2025 all-foreign total of **US$9,270.477 billion** (U.S. Department of the Treasury, 2026a). COFER trends use the **September 30, 2026 release**; the October 2025 Q2 valuation decomposition is a separately dated historical snapshot (International Monetary Fund, 2026a; Kwende et al., 2025).

1.1 DATA SOURCES

The core empirical sources are selected according to the research question rather than according to which series happens to support the de-dollarization narrative.

Analytical purpose	Dataset/source	Frequency / unit	Primary use	Critical limitation
Country Treasury holdings	TIC/CSLT via FRED; historical Major Foreign Holders table as cross-check	Monthly; US$ billions/millions	RQ1 country trends	Country attribution can reflect custodial location rather than ultimate beneficial ownership
Country and aggregate Treasury positions, transactions and valuation	TIC/CSLT via FRED and published Kansas City Fed decomposition	Monthly	RQ1–RQ2 holdings-versus-flow analysis	Holdings and transactions are not interchangeable
Global FX reserve currency shares	IMF COFER	Quarterly; % and US$	RQ1 reserve-currency trend	Country submissions are confidential; gold excluded
Gold demand	World Gold Council Gold Demand Trends	Quarterly/annual; tonnes	RQ1 alternative reserve accumulation	Includes estimates for some unreported purchases
Country gold stocks	WGC quarterly workbook; RBI and Bank of Russia disclosures	Quarterly/dated; tonnes or %	RQ1 quantities and custody	Stocks do not identify funding assets; custody and denomination are distinct
Treasury yields	10-year Treasury constant-maturity yield	Daily/monthly; %	RQ2 benchmark rate	Yield is not a total-return index
Portfolio equity return	S&P 500 Total Return Index, or equivalent broad U.S. total-return index	Monthly	RQ3	Licensed data may be required

Analytical purpose	Dataset/source	Frequency / unit	Primary use	Critical limitation
Portfolio fixed-income return	Bloomberg U.S. Aggregate Bond Total Return Index, or a Treasury total-return index if pure sovereign exposure is desired	Monthly	RQ3	Aggregate index includes non-Treasury bonds
Gold return	LBMA Gold Price PM or GLD price/total return proxy	Monthly	RQ3	Physical gold and ETF implementation costs differ
U.S. cash proxy	BIL ETF adjusted monthly return	Monthly	USD Sharpe ratio	An ETF proxy is not guaranteed risk-free; not a MYR cash benchmark

TIC reports end-of-period holdings and transaction information. The analysis uses the **Continuous Securities Long-Term (CSLT)** data, which combine historical estimates with the expanded SLT reports. Through January 2023, transactions and price valuation are estimated using holdings, price indexes and other adjustments; from February 2023 they are directly collected, with other changes residual. CSLT provides a comparable published series across the transition, but the change in measurement remains relevant to inference. Country series include official and private holders (McCallum et al., 2026; U.S. Department of the Treasury, 2026a).

IMF COFER requires a different interpretation. It reports the global currency composition of foreign-exchange reserves, converts positions at end-period exchange rates, excludes monetary gold, and does not release individual-country currency allocations because submissions are confidential. Therefore, **COFER cannot establish that China, India or Brazil individually reduced their dollar reserve share by a particular number of percentage points**. It can only establish the global reserve-currency trend (International Monetary Fund, 2026a; Kwende et al., 2025).

World Gold Council data complement, rather than substitute for, COFER. Gold is outside the COFER foreign-exchange-reserve denominator; therefore, a chart combining “USD share of COFER” with “gold share of total international reserves” without reconciling denominators would be statistically misleading. World Gold Council also notes that its aggregate central-bank demand measure combines reported reserve changes with an estimate for unreported purchases (World Gold Council, 2026a).

Country gold statistics principally draw on IMF International Financial Statistics and central-bank reports. The analysis verifies all five founding members against the World Gold Council quarterly workbook, restricted to **2018 Q4–2025 Q4**. It supplies synchronized year-end comparisons, including India at **600.44 tonnes in 2018 and 880.34 tonnes in 2025**. The RBI custody report separately records **880.18 tonnes at September 2025**; that is not a December observation. The workbook’s original export date is not established (cjseven007, 2026; Reserve Bank of India, 2025; World Gold Council, n.d.).

BIS Project mBridge is relevant only as **institutional context** for payment-system diversification. BIS states that mBridge reached minimum viable product status in mid-2024 and involved the BIS Innovation Hub and the monetary authorities of China/Hong Kong, Thailand and the UAE. It should not be treated as evidence that Treasury reserves were sold, and does not establish transaction volumes for Treasury securities (Bank for International Settlements, 2024).

1.2 DATA VARIABLES & DEFINITIONS

A technically sound report must keep five concepts separate.

Reading the formulas. The symbol Δ means "change in," so ΔH means the change in holdings. Letters such as α and β represent a baseline or the strength of a relationship. Subscripts such as "start" and "end" label a date or period. Each equation summarizes a financial idea; its meaning and limitations are explained beside it.

Treasury holdings. This is the end-period reported value of Treasury securities attributed by TIC to a particular country. Long-term securities are reported at market value; Treasury bills and certificates are at face value. Calculate the change as follows:

$$\Delta H = H_{\text{end}} - H_{\text{start}}$$

Here H means reported holdings. The equation simply compares the ending value with the starting value.

This change is **not automatically a purchase or sale**.

A useful accounting decomposition is:

$$\begin{aligned}\Delta H = F &+ \text{valuation change} + \text{repayment change} \\ &+ \text{custody/classification change} + \text{measurement difference}.\end{aligned}$$

Here F means net purchases. The other components are written out to distinguish them from transaction flows. **The principle is that holdings can change for several reasons besides trading.**

Here, net purchases means purchases minus sales, excluding principal repayments. Price changes mean gains or losses caused by changing security prices. Repayment changes reflect principal paid back to investors; custody or classification changes reflect changes in how holdings are recorded. Measurement differences cover the remaining unexplained amount. If the transaction series already includes principal repayments, include them in net transactions only once. New Treasury issuance changes an investor's position only when acquired; issuance is not an additional position change on top of its purchase flow. The Federal Reserve Bank of Kansas City emphasizes precisely this need to distinguish active reallocation from passive valuation effects (Dilts Stedman, 2025).

Dollar share of foreign-exchange reserves. This is the value of dollar-denominated foreign-exchange reserves divided by the applicable COFER foreign-exchange-reserve total. The October 2025 article uses the then-published allocated-reserves denominator. Starting with the 2025 Q3 release, the IMF eliminated the unallocated category, imputed currency composition, and revised the series back to 2000 Q1. The revised quarterly series therefore refers to total FX reserves including imputed currency allocations. The archived export retains the legacy indicator label "Allocated foreign exchange reserves"; the revised total includes imputed allocations. Gold remains excluded. Because positions are converted into dollars at period-end exchange rates, the percentage can move even without reserve-manager trading (International Monetary Fund, 2026a, n.d.; Kwende et al., 2025).

Transaction flow (net purchases). This is net buying or selling over a period. In CSLT, it is reported or estimated according to the observation date, not always directly observed. For the long-term calculation used here:

$$\Delta H = F + \text{valuation change} + \text{other changes}$$

Here F means net transactions. These are positive for net buying and negative for net selling; the final category retains the other recorded changes. Do not add a separate redemption or custody term on top of this three-component identity. Country flows are not identified sovereign flows. They improve the description of demand but do not alone identify an exogenous yield shock (McCallum et al., 2026).

Gold reserves. This is reported monetary-gold holdings in tonnes for a country at the stated observation date. A change in tonnes measures physical accumulation independently of the gold price, whereas a change in gold's dollar value or reserve share also reflects valuation. Physical accumulation does not itself identify active sales of another reserve asset (Bertaut et al., 2025; World Gold Council, n.d.).

Portfolio total return. This includes price changes plus distributed income such as coupons and dividends. A Treasury **yield series is not a bond-return series**. Using DGS10 or another 10-year yield as the "return on Treasuries" would be an elementary measurement error because higher yields can coincide with negative bond price returns.

1.3 ANALYTICAL METHODS

Trend and relative-holdings analysis

For each country, absolute change compares the ending and starting holdings, using the ΔH definition above. Percentage change puts that movement in proportion to the starting value:

$$\text{Percentage change} = \left(\frac{H_{2025}}{H_{2018}} - 1\right) \times 100.$$

A negative answer means the holdings fell. The dates must match the comparison being discussed.

A second normalization asks whether the country became less important relative to the entire foreign investor base:

$$\text{Country share } (\%) = \frac{H_{\text{country}}}{H_{\text{all foreign}}} \times 100$$

Divide the country's holdings by total foreign holdings measured on the same date.

The year-end TIC observations produce the following comparison.

The eight December observations underlying the existing country charts are also reported explicitly:

Year	China	India	Brazil
2018	1,124.3	141.3	303.1
2019	1,069.9	162.0	281.8
2020	1,072.3	216.2	258.3
2021	1,040.3	198.9	240.6
2022	867.1	224.1	217.4
2023	816.3	233.7	230.3

Year	China	India	Brazil
2024	759.0	219.1	201.6
2025	684.4	182.9	168.7

US$ billions, December positions; (U.S. Department of the Treasury, 2026a); historical TIC tables provide a cross-check.

Country	Dec. 2018, US$ bn	Dec. 2025, US$ bn	Absolute change	Relative change	Share of all foreign holdings, 2018	Share, 2025
China	1,124.3	684.4	−439.9	−39.1%	17.93%	7.38%
India	141.3	182.9	+41.6	+29.4%	2.25%	1.97%
Brazil	303.1	168.7	−134.4	−44.3%	4.83%	1.82%
China + India + Brazil	1,568.7	1,036.0	−532.7	−34.0%	25.02%	11.18%
All foreign holders	6,270.1	9,270.5	+3,000.3	+47.9%	100%	100%

Author calculations from TIC/CSLT FRED year-end holdings; historical TIC values provide a cross-check. These are country-attributed reported positions, not proven central-bank transaction flows; Treasury bills and certificates are at face value. Changes are calculated before display rounding. (U.S. Department of the Treasury, 2026a).

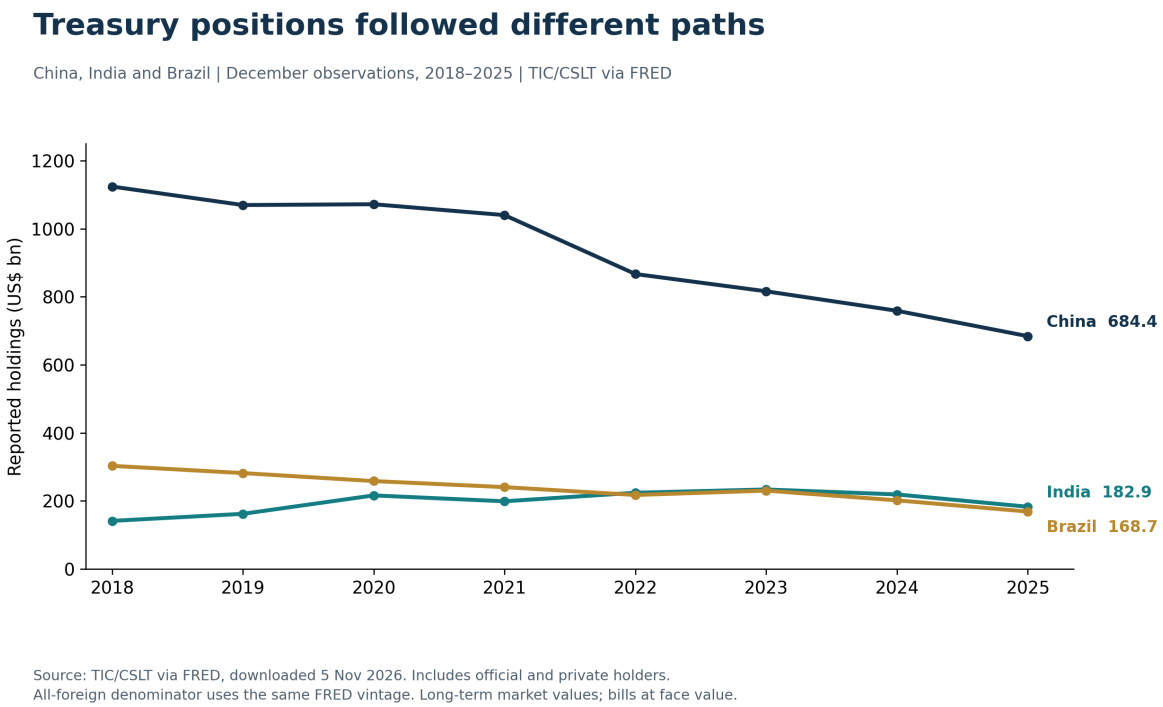


Figure 1. Reported year-end Treasury positions for China, India and Brazil, 2018–2025. Author chart from (U.S. Department of the Treasury, 2026a); all eight December observations are preserved. Positions include official and private holders and are not transaction flows.

The absolute chart reveals a clear secular reduction in China's and Brazil's reported positions. India's pattern is different: holdings rose substantially through the middle of the sample before declining in 2024–2025, finishing above their 2018 level. The most important analytical implication is that **BRICS should not be modeled as one portfolio manager** (U.S. Department of the Treasury, 2026c).

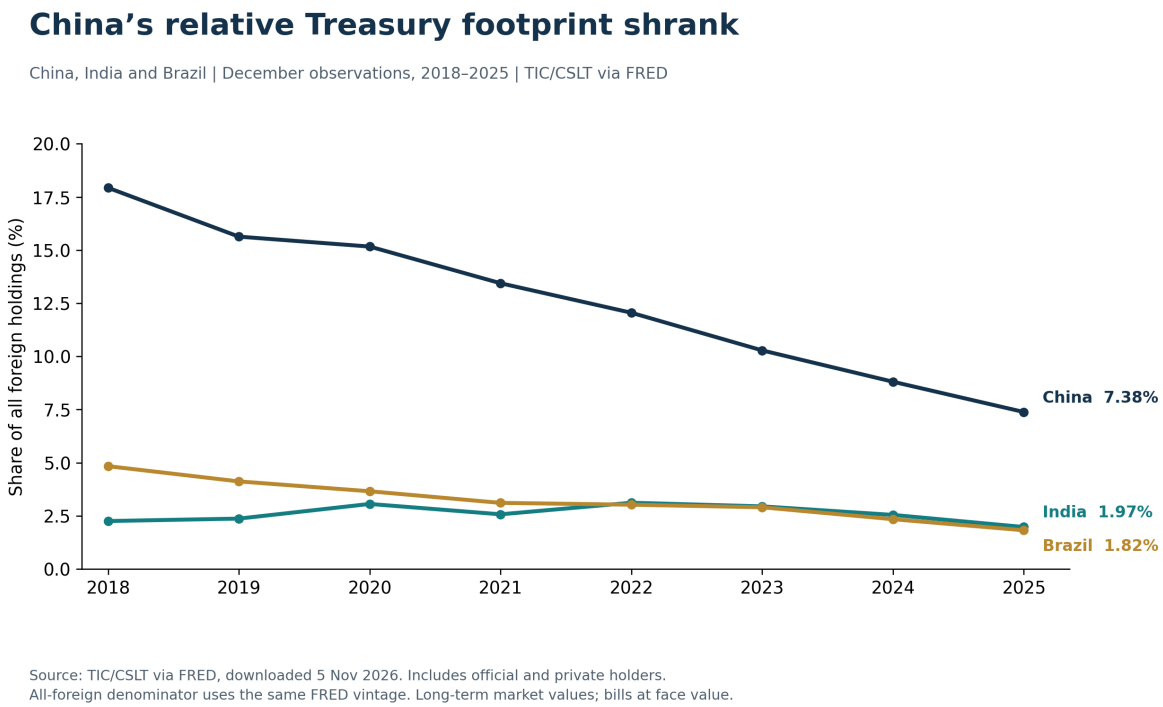


Figure 2. Country positions divided by all foreign-held Treasuries, using the TIC/CSLT FRED vintage. Author calculations from (U.S. Department of the Treasury, 2026a). The denominator is neither total U.S. government debt nor global FX reserves.

The relative chart is arguably more informative. China's position declined not only in dollars but from approximately 17.9% to 7.4% of total foreign Treasury holdings. Brazil also lost relative weight. India's dollar holdings rose, but its share of all foreign Treasury holdings fell modestly because the total foreign investor base expanded much faster. Thus, “de-dollarization” can mean at least three different things: selling existing dollar securities, buying fewer new dollar securities than in the past, or allowing dollar assets to shrink as a share of a growing reserve portfolio. Those mechanisms have different implications for Treasury-market pressure.

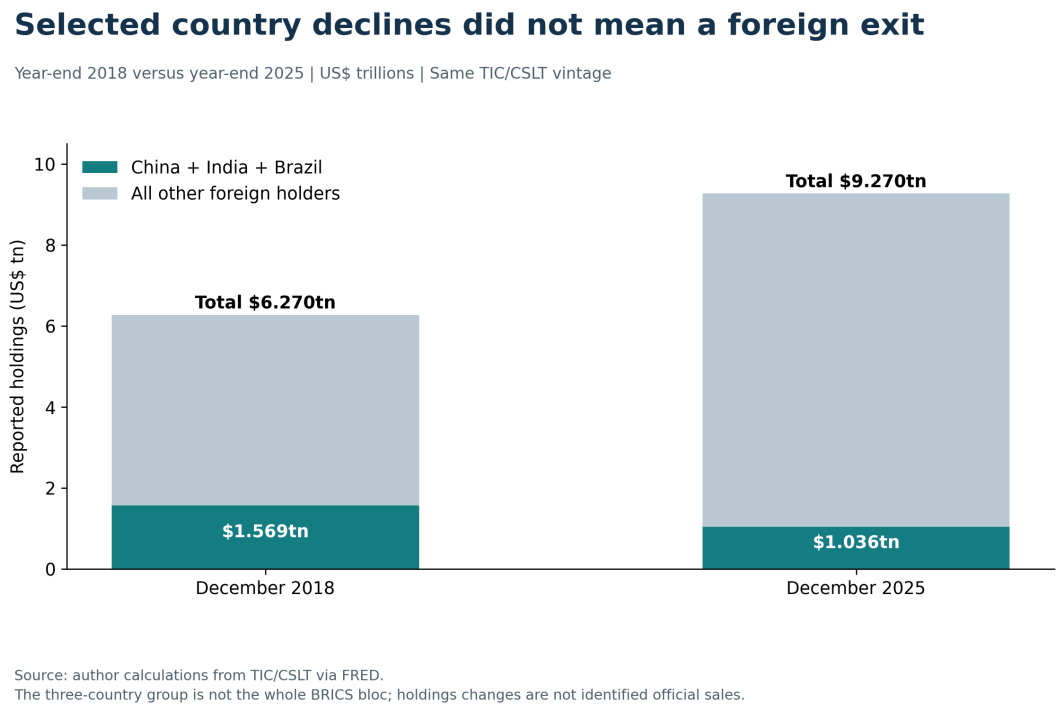


Figure 3. Selected country reductions occurred while the broader foreign investor base expanded. Author calculations from (U.S. Department of the Treasury, 2026a). “All other foreign holders” is the residual after subtracting the three displayed countries.

Russia and South Africa are quantitatively smaller Treasury holders. TIC records show Russia at US$13.2 billion in December 2018; its reported position subsequently fell to extremely low levels. South Africa was at US$10.5 billion in December 2018 and US$13.8 billion at the end of 2019. Because these positions are small compared with China and because detailed lower-ranked country coverage is handled differently across historical TIC tables, Russia and South Africa should be reported separately rather than mechanically aggregated with the three large holders (U.S. Department of the Treasury, 2026c).

Reserve-share decomposition and gold

At the aggregate reserve-system level, the dollar remained dominant at 56.52% of COFER foreign-exchange reserves at the end of 2025. That share is lower than earlier decades, and the IMF literature describes the long-run decline as gradual diversification into a wider set of currencies rather than a simple dollar-to-renminbi replacement. Arslanalp et al. (2022) emphasize the role of nontraditional reserve currencies in this process (Arslanalp et al., 2022; International Monetary Fund, 2026a).

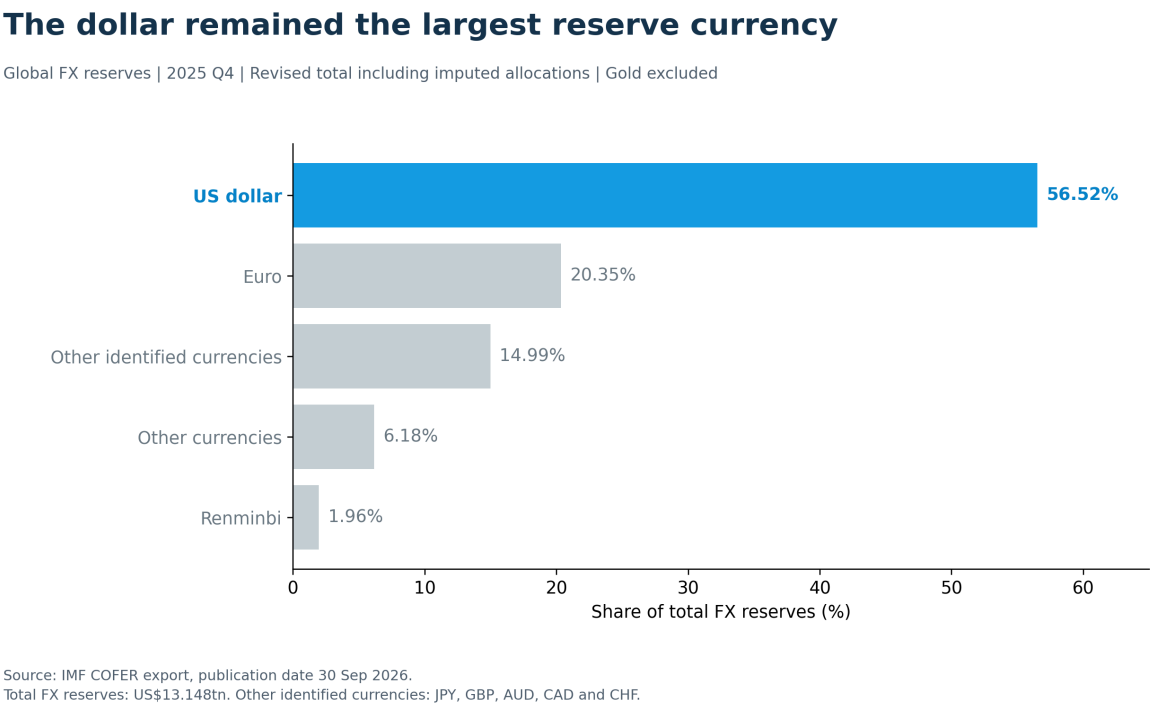


Figure 4. Global FX reserve currency composition, 2025 Q4. Author chart from the September 30, 2026 IMF COFER release. The U.S. dollar is highlighted in blue; the other categories are grey. Other identified currencies combine yen, sterling, Australian dollar, Canadian dollar and Swiss franc. Includes imputed currency allocations and excludes gold. This is not a BRICS-country breakdown (International Monetary Fund, 2026a).

A useful decomposition is:

$$\Delta s = \text{transaction effect} + \text{FX valuation effect} + \text{other valuation effects} + e.$$

Here s is the dollar's reserve share, FX means foreign exchange, and e represents remaining measurement differences. **A falling share can reflect valuation rather than selling.**

Each contribution is measured in percentage points. This is a way to organize the change in a share; it does not trace money between currencies.

The importance of this decomposition is demonstrated by the **October 1, 2025 publication snapshot for 2025 Q2**, before the subsequent COFER imputation revision. The raw dollar share fell from 57.79% to 56.32%, but at unchanged exchange rates it would have been 57.67%. Thus, approximately 92% of the headline decline reflected currency valuation. The remaining 0.12 percentage points are a non-FX remainder, not an identified measure of active dollar sales (Kwende et al., 2025).

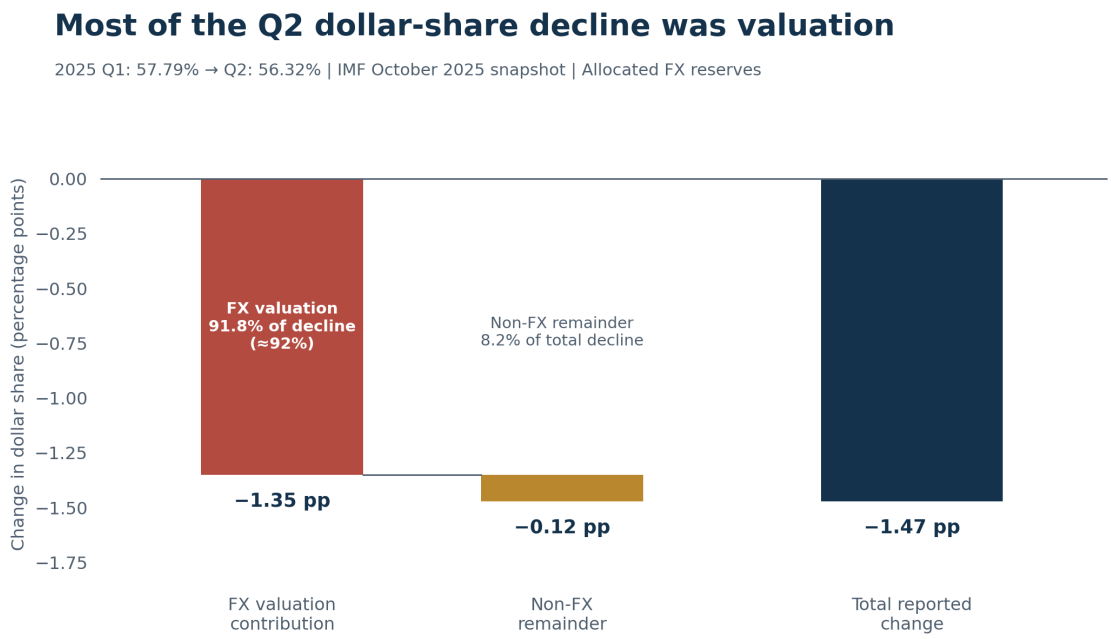


Figure 5. Zero-based change decomposition of the October 2025 IMF snapshot: FX valuation contributes −1.35 percentage points and the non-FX remainder −0.12 points, summing to a −1.47-point total. The reported share moves from 57.79% to 56.32%. The separate total bar is a sum, not a third contribution; the connecting line joins the first contribution to the second. The contributions are an accounting decomposition, not a sequence of trades. The non-FX remainder is not identified active dollar sales. Author calculation from (Kwende et al., 2025); this older allocated-reserves vintage is separate from Figure 4.

This single observation rejects a crude sales interpretation; the valid inference is:

A lower dollar reserve share does not, by itself, show an equivalent amount of dollar-asset sales.

A lower share can have several causes, so the transactions must be checked separately.

Gold nevertheless provides evidence of genuine reserve diversification at the global level. Central banks bought an estimated 863 tonnes in 2025, below the 1,000-tonne threshold exceeded during the previous three years but still historically elevated. The World Gold Council's 2025 survey reported that 95% of respondents expected global official gold reserves to rise over the following year, and 43% expected their own holdings to rise. Survey expectations should be treated as sentiment evidence, not as completed transaction data (World Gold Council, 2025, 2026a).

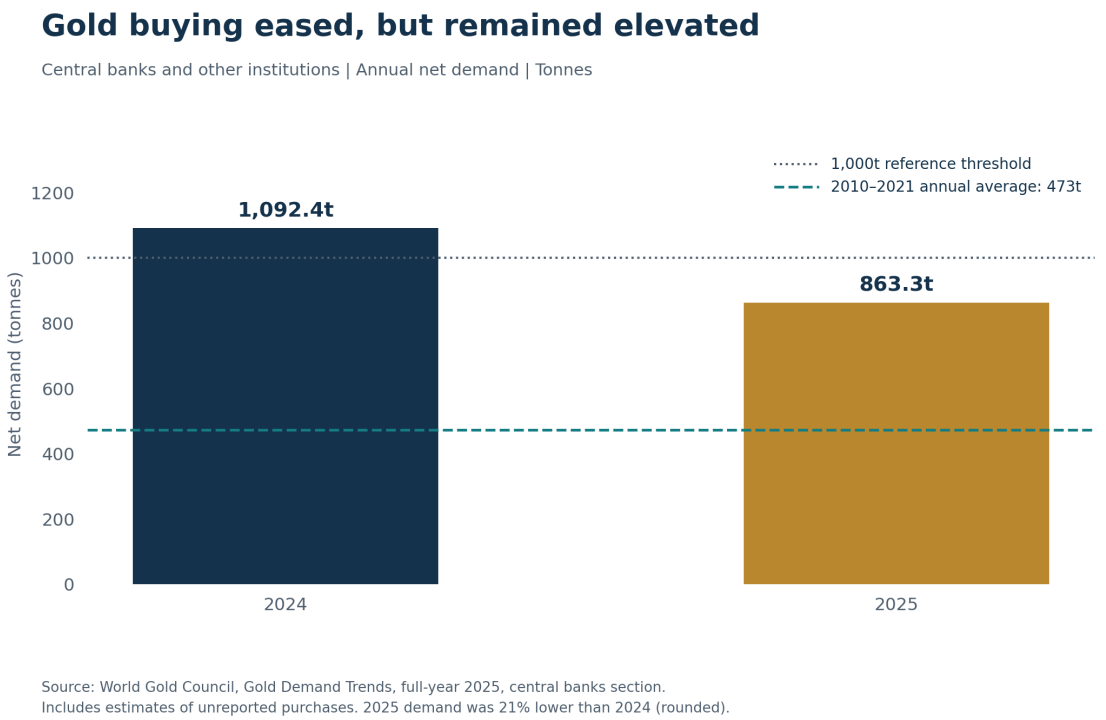


Figure 6. Annual gold demand, central banks and other institutions: 1,092.4 tonnes in 2024 and 863.3 tonnes in 2025. Author chart from (World Gold Council, 2026a); the same report supplies the 473-tonne 2010–2021 annual average. Both reference lines are explicitly identified: the dotted gray line is a 1,000-tonne comparison threshold and the dashed teal line is the historical annual average, not 2025 demand. Bar labels show the published annual totals. Demand includes estimated unreported purchases.

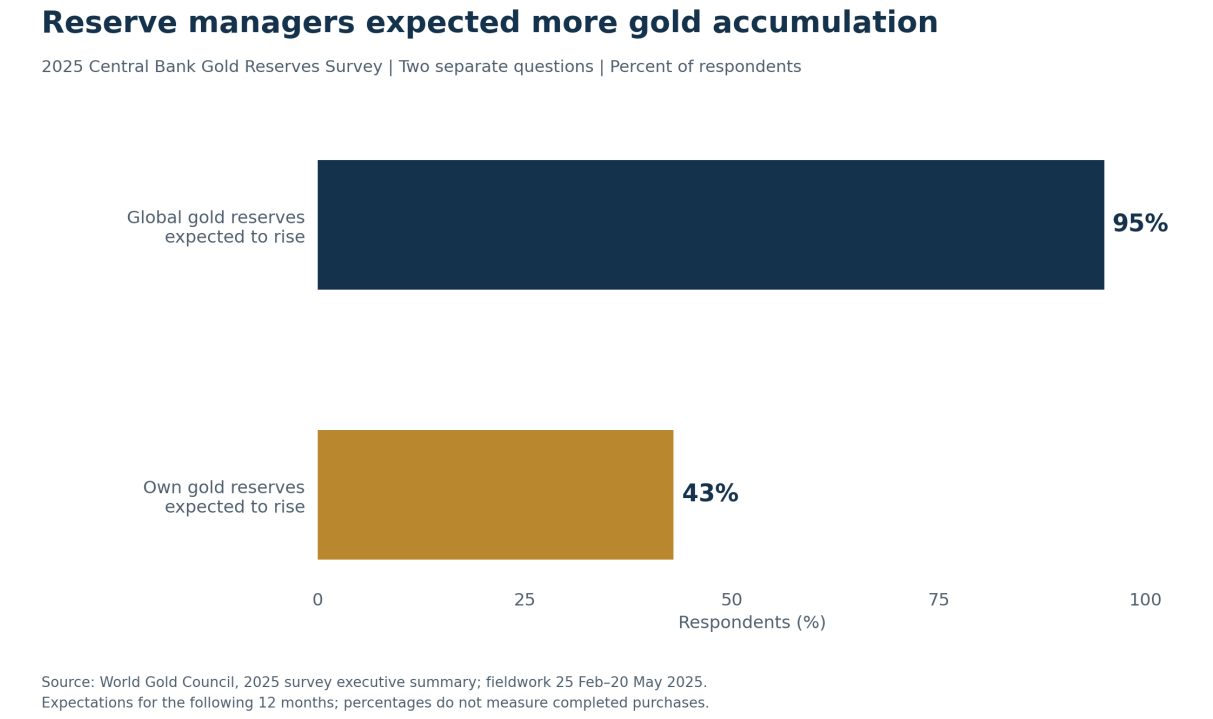


Figure 7. Two separate forward-looking questions in the (World Gold Council, 2025). These percentages describe respondents' expectations for the following 12 months, not the percentage of all central banks that actually bought gold.

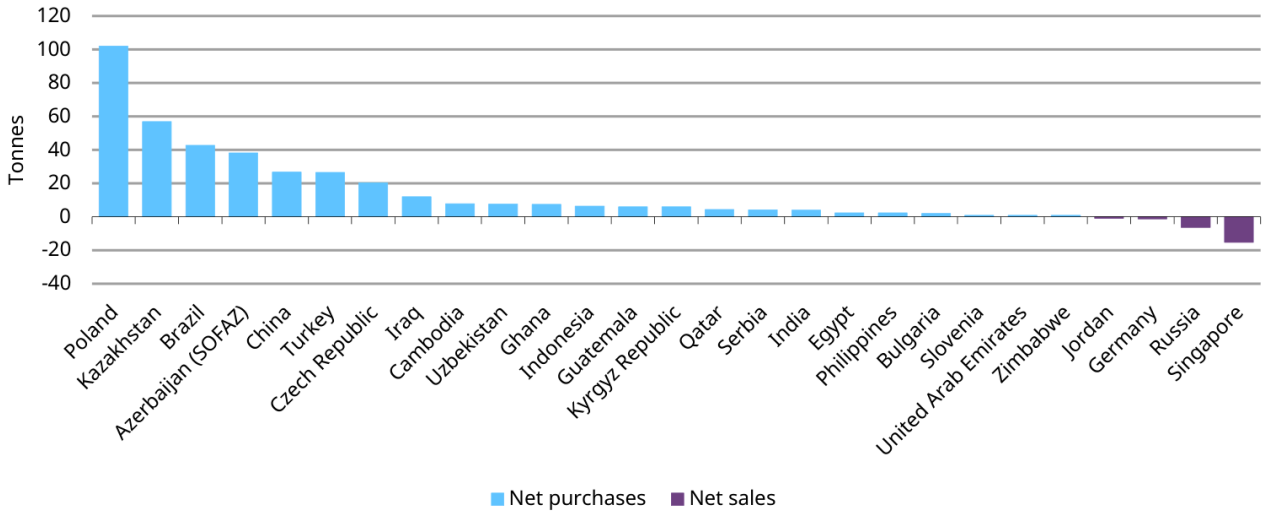
Country-level gold evidence and reporting dates. The preserved WGC quarterly workbook now permits synchronized end-2018/end-2025 comparisons for all five founding members. Its end-2018 China observation is 1,852.51 tonnes, a small revision from the 1,852.2 tonnes quoted in the January 2019 publication. These vintages are not mixed in the growth calculations. India's WGC year-end observations differ in date from the RBI custody disclosures (Reserve Bank of India, 2025; World Gold Council, n.d., 2019).

Country	Treasury evidence, December 2018 → December 2025	Gold, end-2018 → end-2025 (tonnes)	Change (tonnes)	Interpretation
China	1,124.3 → 684.4 bn	1,852.51 → 2,306.30	+453.79	Lower attributed Treasury exposure alongside reported gold accumulation
India	141.3 → 182.9 bn	600.44 → 880.34	+279.90	Both exposures ended above their 2018 baseline
Brazil	303.1 → 168.7 bn	67.36 → 172.44	+105.08	Gold gains concentrated in 2021 and late 2025; funding not identified
Russia	13.216 → 0.029 bn	2,113.03 → 2,326.52	+213.49	Large gold stock and near-zero directly reported Treasury position
South Africa	Smaller Treasury holder; excluded from new four-country aggregate	125.31 → 125.50	+0.19	Reported gold quantity was approximately stable

Treasury values are US$ billions and include official and private holders. Gold quantities come from the same preserved WGC workbook. Physical stock changes do not identify Treasury-sale proceeds or a change in total dollar assets. Detailed trajectories and dates appear in RQ1 (U.S. Department of the Treasury, 2026a; World Gold Council, n.d.).

Chart 13: The National Bank of Poland topped global gold purchases in 2025

Reported 2025 central bank gold demand, tonnes*



*Data to 31 December 2025 where available.
Source: IMF, respective central banks, World Gold Council

*Figure 8. Extracted original Chart 13, (World Gold Council, 2026a, p. 17), with title, axes, legend, reporting-date note and source preserved. Its reported country flows differ from the aggregate demand estimate, which includes unreported buying. SOFAZ is a sovereign wealth fund; some country observations end before December. *

Quantity versus reserve value. More gold in tonnes is evidence of physical accumulation, but it does not reveal which asset financed it or establish a fall in the country's dollar share. Bertaut et al. (2025) add an important qualification: the sharp rise in gold's market-value share of global reserves since 2015 mostly reflects higher gold prices, while physical holdings rose much less. Their note uses an earlier data vintage and is not a December 2025 estimate (Bertaut et al., 2025).

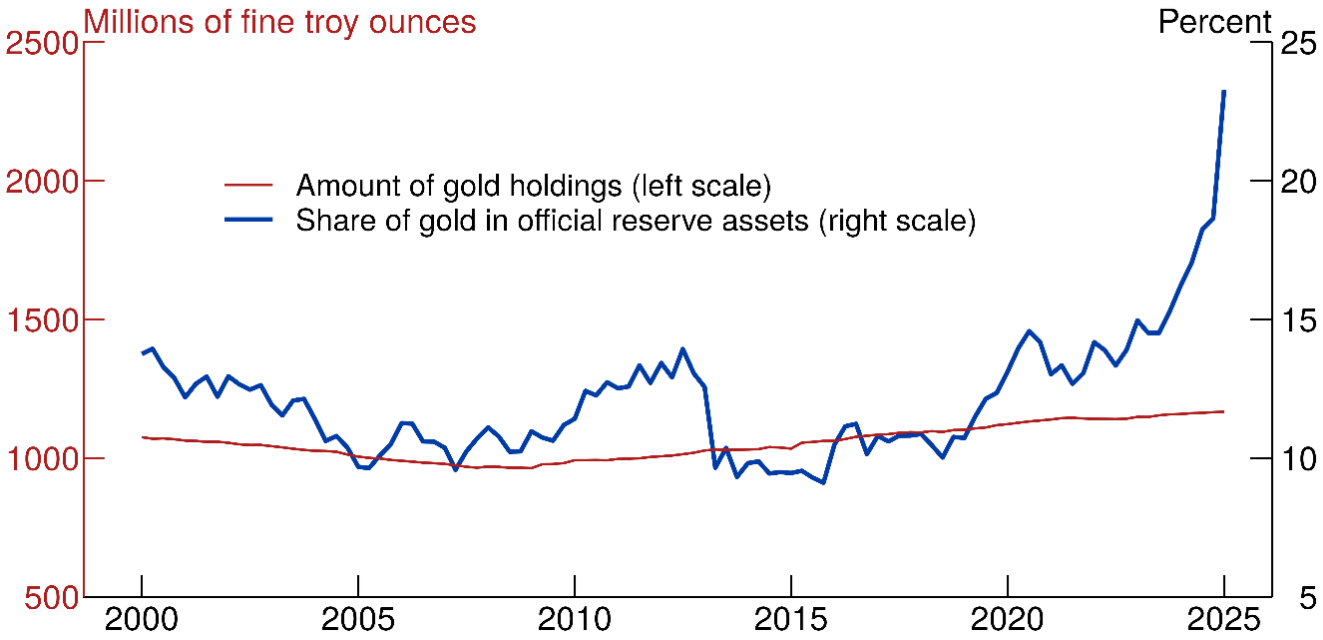


Figure 9. Original Figure 3 from (Bertaut et al., 2025), from the Federal Reserve. The blue series is gold's market-value share of official reserves, in percent on the right axis; the red series is physical holdings in millions of fine troy ounces on the left axis. The series have different units and the chart is contextual evidence through the source's publication, not the core sample's year-end gold-growth estimate.

1.4 FORMULAS, METRICS & ASSUMPTIONS

Fixed-income valuation and stress testing

The price of a fixed-rate bond is the discounted value of coupons and principal:

$$P = \mathrm{PV}_1 + \mathrm{PV}_2 + \cdots + \mathrm{PV}_n$$

P = bond price; $\mathrm{PV}_1, \mathrm{PV}_2, \ldots, \mathrm{PV}_n$ = the present values of its future payments. **PV means present value throughout this report.** Add every discounted coupon and the principal repayment.

For each payment:

$$\mathrm{PV} = \frac{\mathrm{CF}}{(1+r)^n}$$

PV = present value; CF = the future cash flow being valued; r = interest rate per payment period as a decimal; n = number of periods until that payment.

A coupon is the bond's regular interest payment; principal is the amount repaid at maturity. The interest rate per payment period equals the quoted annual yield, as a decimal, divided by the number of payments per year. Add the discounted value of every coupon and the principal repayment. For example, a single payment of 105 due in one year is worth $105 \div 1.05 = 100$ today at a 5% yearly yield. This method assumes a fixed-rate bond with no embedded options, valued on a coupon-payment date.

Duration measures how sensitive a bond's price is to a change in yield. A modified duration of 7.5 means that a 1-percentage-point rise in yield produces about a 7.5% price fall in this simple estimate. The calculation is:

$$\text{Price change } (\%) \approx -D \times \frac{\Delta y_{\text{bp}}}{100}$$

Here D is modified duration and Δy_{bp} is the yield change in basis points. The sign $\approx$ means "approximately equal to." **Higher yields generally reduce the price of an existing fixed-rate bond.**

A more detailed calculation would allow for **convexity**, the curvature in the price–yield relationship. Duration gives a straight-line approximation, so it becomes less accurate for larger yield changes. The stress tables below use the simpler duration estimate and exclude that curvature adjustment.

For an assumed modified duration of 7.5, the first-order stress results are:

Parallel yield shock	Approximate price change
+25 bp	−1.88%
+50 bp	−3.75%
+57 bp	−4.28%
+75 bp	−5.63%
+100 bp	−7.50%
+150 bp	−11.25%

These calculations isolate interest-rate sensitivity only. They exclude coupon income, roll-down, convexity, reinvestment, changing spreads and nonparallel yield-curve movements. Therefore, they are **stress tests, not forecasts**.

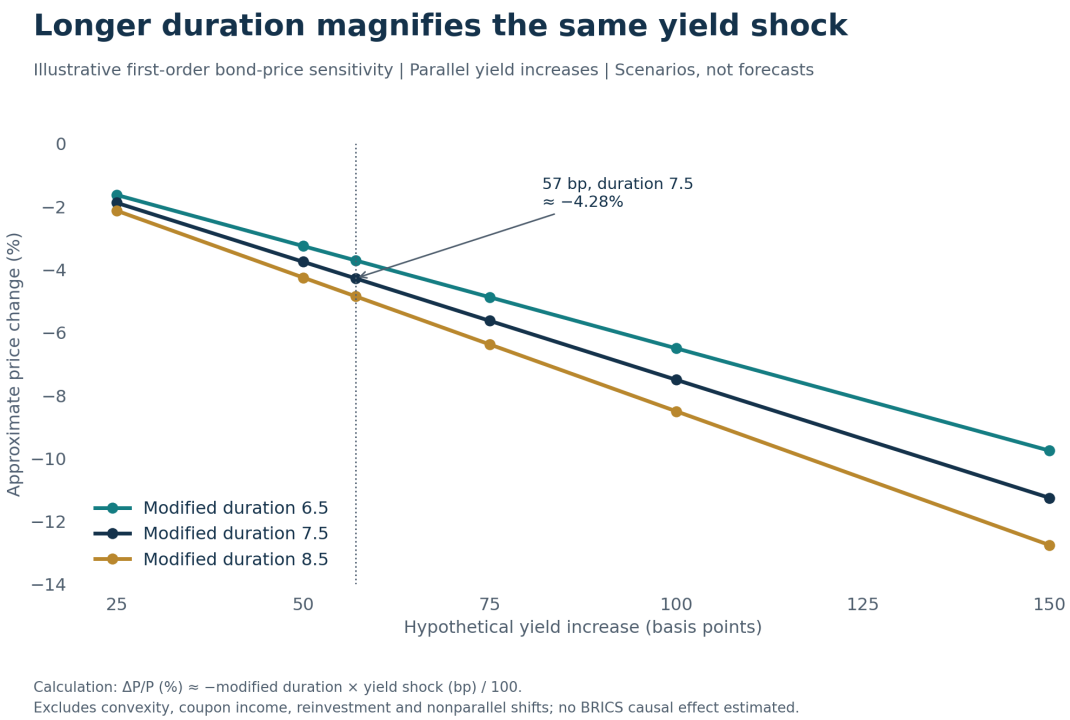


Figure 10. Author-modeled first-order sensitivity for modified durations 6.5, 7.5 and 8.5. Price change (%) = −duration × yield shock (bp) / 100. The 57bp annotation is a hypothetical yield input; no BRICS-specific loss is estimated. Convexity, coupon income and reinvestment are excluded.

Figure 38 summarizes the conditional transmission mechanism. Federal Reserve policy, inflation, issuance and growth remain competing determinants; the figure does not attribute a yield change to BRICS.

Illustrative econometric specification and identification limits

Status of the equation. This illustrative regression expresses a possible association; it is not estimated in this report and does not establish a BRICS effect.

Economic purpose. The proposed hypothesis is that lower foreign-official net purchases, relative to Treasury supply and other investors' demand, could lower existing bond prices and raise yields. Other investors may offset the change, so this is a conditional mechanism. A regression would attempt to distinguish that association from changes in monetary-policy expectations, inflation expectations, issuance and risk compensation. One illustrative specification is:

$$\Delta y = \alpha + \beta F + \text{other influences} + e.$$

Here Δy is the monthly change in the 10-year Treasury yield, F is net official Treasury purchases, α is a baseline, β measures the association with purchases, and e is the unexplained part. **The question is whether official buying or selling helps explain yield changes after allowing for other influences.**

A “contribution” means the amount associated with that factor after accounting for the others. These contributions would have to be estimated from data; the formula does not mean that the raw figures can simply be added together. For example, Treasury purchases measured in dollars must first be converted into an estimated yield contribution measured in basis points.

Intended variables, not an implemented dataset. For a hypothetical monthly model, the variables would require the following definitions before estimation.

Plain-language term	Intended meaning and measurement	Purpose
Yield change	Monthly change in the 10-year Treasury yield, in basis points	Outcome to be explained; a yield change is not a bond total return
Net official purchases	Monthly net Treasury purchases by foreign official institutions, for example in US$ billions; purchases positive and sales negative	Candidate demand measure; it must be transactions, not the change in market-value holdings
Change in expected inflation	Change in a specified measure of expected inflation over a fixed horizon, in percentage points	Inflation expectations can affect nominal yields; the horizon and data source must be defined
Policy-rate change	Change in a specified policy-rate measure, in percentage points	A rough monetary-policy control; the current policy rate alone does not capture the expected future rate path
New Treasury supply	A precisely defined supply measure, such as monthly net marketable Treasury issuance in US$ billions	Treasury supply may affect the yield required by marginal buyers; gross issuance, net issuance and duration-weighted supply are different measures
Term premium	A model-estimated 10-year term premium, in basis points	Intended to represent compensation for holding long-duration risk; its role as a control is problematic for this question
Baseline change	Starting estimate of the monthly yield change, in basis points	The model's starting point before the other contributions are added
Estimated strength of each relationship	Numbers that turn each input into a yield contribution	These would be calculated from data; they are not known in this report
Unexplained change	Unexplained yield change, in basis points	Omitted influences and measurement error

The hypothesis is: **greater official buying may be associated with lower yields when the other factors are held constant**. The strength of this relationship could be reported as the yield change, in basis points, associated with each additional US$1 billion of net purchases. **The report has not estimated either the direction or the size of that relationship.**

Why this is not a ready-to-use causal model. First, aggregate foreign-official flows include economies outside BRICS. A coefficient on worldwide purchases cannot automatically identify BRICS de-dollarization, and country-attributed holdings do not isolate central-bank transactions.

Second, the term premium is itself an estimated component of a long-term yield. Federal Reserve documentation defines it relative to the expected average short rate, and the New York Fed estimates it from Treasury yields. A contemporaneous term-premium control could therefore absorb the very channel through which a demand shock affects yields, or introduce mechanical dependence on the yield being explained. This is a potential **overcontrol and endogeneity problem**, rather than a rule that every regression including a term premium is invalid. A model targeting the total demand effect would need to justify excluding such a mediator or analyze that channel separately (Board of Governors of the Federal Reserve System, n.d.; Federal Reserve Bank of New York, n.d.).

Third, timing, scaling and transformations have not been justified. The illustration mixes monthly changes, transaction flows and a term-premium level. Different regressor transformations are not automatically mathematically wrong, but frequency alignment, lags, stationarity, persistent series and the choice between term-premium levels and changes would need explicit justification and diagnostics. Inflation expectations and the policy-rate proxy also need a clear role because both can overlap with the expected short-rate path priced into nominal yields.

Fourth, **reverse causality and simultaneity** remain: official investors may react to yields, while inflation news, policy announcements, issuance and risk sentiment can move flows and yields together. Custodial attribution and security-price valuation add measurement error if country holdings are substituted for transactions. The February 2023 change in TIC transaction reporting also needs treatment in a 2018–2025 flow panel. None of these issues is resolved simply by adding controls (Dilts Stedman, 2025; U.S. Department of the Treasury, n.d.).

In this notation, $\beta < 0$ would mean that more official buying is associated with lower yields, after accounting for the included influences. **This report has not estimated β or established that relationship for BRICS.**

The other influences must be justified and measured over matching periods. Expressing purchases as a share of the Treasury market can help compare markets of different sizes, but it cannot establish what caused a yield change or isolate BRICS activity from worldwide official purchases.

Ahmed and Rebucci (2024) identify flow shocks using changes in volatility in a structural VAR; their results depend on identifying assumptions and the response horizon. Wolcott (2020) combines a structural VAR with an affine term-structure model and finds a stronger effect on medium-term yields. The papers strengthen the methodological case for careful identification; neither supplies a universal coefficient that can be applied to the multi-year decline in China's TIC position. (Ahmed & Rebucci, 2024; Wolcott, 2020).

Analysis used in this report. The study combines descriptive Treasury and reserve data, published yield estimates, duration scenarios and an ETF portfolio analysis. It does not estimate a yield/flow regression or a BRICS-specific coefficient. A yield change in basis points can be calculated as:

$$\Delta y_{\text{bp}} = (y_{\text{end}} - y_{\text{start}}) \times 100$$

Here y_{end} and y_{start} are quoted yields in percent units.

Here the quoted yields are in percent units: 3.5 to 4.0 implies a 50-basis-point rise. This is a unit-conversion example, not a measured change attributed to BRICS. Comparing two observed trends would still not establish causation.

The duration relationship above provides the investment-risk bridge.

Use basis points in this version of the formula: $1 \text{ basis point} = 0.01 \text{ percentage points}$. As a separate illustration, an assumed modified duration of 8 years and a 50-basis-point increase give $-8 \times 50 \div 100 = -4\%$, or approximately a 4% price decline. On an assumed USD 1 million position, that is approximately USD 40,000 of price loss before coupon income, convexity and other effects. These are hypothetical inputs; the main scenario table and Figure 10 use their explicitly stated duration assumptions. No observed BRICS-caused loss is inferred.

For RQ3, the published State Street simulation remains the principal literature benchmark. A separate 2018–2025 ETF backtest and correlation matrix are now computed as sample-specific robustness evidence. A future regression would require an identification strategy, estimation and diagnostics; no new regression is estimated here.

Portfolio backtest framework

The original-sample portfolio comparison uses 96 aligned monthly **total-return proxies** from January 2018 through December 2025 and annual rebalancing, alongside the longer published State Street benchmark.

Portfolio	Equities	Bonds	Gold	Interpretation
Baseline	60%	40%	0%	Conventional 60/40
Gold-5 even funded	57.5%	37.5%	5%	Gold funded equally from stocks and bonds
Gold-10 even funded	55%	35%	10%	Larger diversified gold sleeve
Gold-5 bond funded	60%	35%	5%	Gold substitutes only for fixed income
Gold-10 bond funded	60%	30%	10%	Strongest test of “gold replacing bonds”

These definitions parallel a transparent State Street methodology and make the funding decision explicit. That decision matters: a gold allocation funded from bonds is not the same portfolio experiment as one funded equally from stocks and bonds (Doshi et al., 2026).

Monthly portfolio return is:

$$R_P = w_S \times R_S + w_B \times R_B + w_G \times R_G$$

R_P = portfolio return; w = portfolio weight; R = asset return. The labels S, B and G mean stocks, bonds and gold.

Use the weights at the start of the month, expressed as decimals: 60% becomes 0.60. For example, if stocks return 2%, bonds 1% and gold 0% in a 60/35/5 portfolio, the monthly return is $0.60 \times 2\% + 0.35 \times 1\% + 0.05 \times 0\% = 1.55\%$.

Annualized arithmetic return can be reported as:

$$\text{Annual average return} = 12 \times \text{average monthly return}$$

This is an arithmetic average scaled to one year. It does not include growth on earlier gains.

Compound annual growth rate, which includes growth on earlier gains, can be calculated as:

$$\text{CAGR}(\%) = \left[\left(\frac{V_{\text{end}}}{V_{\text{start}}}\right)^{1/T} - 1\right] \times 100$$

V_{end} = ending portfolio value; V_{start} = starting portfolio value; T = number of years. CAGR means compound annual growth rate.

This assumes income is reinvested and there are no outside deposits or withdrawals. To obtain the ending value from monthly returns, multiply the starting value by $(1 + R_1)$, then by $(1 + R_2)$, and continue for every month. Here R_1 and R_2 are the first and second monthly returns. Use decimal returns in these growth factors.

Annualized volatility is:

$$v_{\text{year}} = v_{\text{month}} \sqrt{12}$$

Here v means volatility. The conversion assumes that monthly returns have negligible serial correlation.

The square root of 12 is about **3.46**. Volatility means the standard deviation of returns: a measure of how widely returns vary around their average.

The Sharpe ratio is:

$$E = R_P - R_F$$

Here E is the extra return above Treasury bills, using returns from the same month.

$$S_{\text{year}} = \sqrt{12} \times \frac{\text{average monthly } E}{\text{standard deviation of monthly } E}$$

Here S is the Sharpe ratio. **It compares extra return with the variability of that extra return.** A higher value is better within this measure, but does not guarantee a smaller drawdown.

Use the average and sample standard deviation of excess returns from the same months. Return and Treasury-bill figures must use the same units. The square-root-of-time annualization assumes negligible serial correlation; material autocorrelation requires a different adjustment. The original ETF exercise uses BIL total returns as a U.S. cash proxy; BIL is not guaranteed risk-free. This convention does not recalculate State Street's published Sharpe ratios.

Maximum drawdown is:

$$\text{Drawdown } (\%) = \left(\frac{V}{V_{\text{peak}}} - 1 \right) \times 100$$

Here V is the current portfolio value and V_{peak} is its highest value so far. **Drawdown asks how far the portfolio has fallen from its previous best level.**

Maximum drawdown is the largest peak-to-trough decline over the sample, reported here as the most negative drawdown percentage.

For example, a fall from a previous peak of 100 to 80 is $(80 \div 100 - 1) \times 100 = -20\%$.

Diversification depends on how assets move together, as well as their individual risk. The three-asset calculation can be kept to ordinary algebra by writing each weighted volatility as A, B or C:

$$A = w_S v_S, \qquad B = w_B v_B, \qquad C = w_G v_G.$$

Here w means portfolio weight, v means volatility, and the labels S, B and G mean stocks, bonds and gold. Then:

$$v_P^2 = A^2 + B^2 + C^2 + 2ABr_{SB} + 2ACr_{SG} + 2BCr_{BG}.$$

The letter r represents the correlation of the labelled pair of assets. Take the square root to obtain portfolio volatility, v_P. Use decimal weights and matching periods and units for all returns.

The principle matters more than the expansion: assets that tend to fall together offer less diversification. This is the three-asset calculation in Markowitz's framework (Markowitz, 1952, pp. 77–91).

The calculated correlation matrix uses the **same 96 aligned monthly total-return observations**, January 2018–December 2025. SPY, AGG and GLD proxy stocks, aggregate bonds and gold:

Monthly return correlation	Equity	Bonds	Gold
Equity	1.000	0.422	0.138
Bonds	0.422	1.000	0.395
Gold	0.138	0.395	1.000

These are descriptive sample correlations, not stable parameters or causal effects. The SPY–AGG correlation was 0.751 in 2022–2023 and −0.222 in 2018–2019; each subperiod contains 24 months (Figure 34). Peer-reviewed evidence nevertheless supports the hypothesis that gold's relationship with stocks is regime-dependent: Baur and Lucey (2010, pp. 217–229) found gold to act as a hedge on average and a safe haven in certain extreme equity-market conditions, while also finding that the safe-haven effect can be short-lived (Baur & Lucey, 2010).

The completed exercise also substitutes IEF for AGG, compares both funding methods and calculates MYR outcomes. Further sensitivity tests could examine:

Sensitivity test	Base case	Alternative
Gold weight	5%, 10%	2%, 15%
Funding source	Even funded	Bond only
Rebalancing	Annual	Quarterly / no rebalancing
Sample	2018–2025	2020–2025; 2022–2025
Risk-free proxy	3-month T-bill	Zero-rate comparison
Bond exposure	Aggregate bonds	7–10-year Treasuries
Gold implementation	LBMA/physical-price proxy	GLD
Return convention	Arithmetic + CAGR	Log-return robustness check

The purpose is not to hunt for the allocation with the highest historical Sharpe ratio. It is to determine whether the conclusion—"gold improves portfolio efficiency"—survives reasonable changes in assumptions.

1.5 LIMITATIONS

The country-level analysis is descriptive and does not establish causation. Pre-February-2023 CSLT flows and valuation changes are estimated; country series combine official and private investors, and the calculated residual is not assigned to a particular motive. The WGC workbook and IMF export were verified internally against preserved source bytes; they were not independently re-downloaded from authenticated publisher databases. Their archival provenance and observation windows are explicit. Differences in country, maturity, date and publication vintage prevent direct substitution of these series for identified sovereign demand.

The largest limitation is **data observability**. TIC country positions combine official and private investors and can be assigned to a custodian's location; COFER does not disclose individual-country currency allocations; and gold reporting dates differ across institutions. A physical gold increase does not identify its funding source (International Monetary Fund, n.d.; U.S. Department of the Treasury, n.d.; World Gold Council, n.d.).

The sample also includes the pandemic, inflation, Federal Reserve tightening, large Treasury issuance and geopolitical shocks. These can move yields independently of reserve allocation, while valuation changes can move reported positions without trading. Accordingly, correlation between China's holdings and U.S. yields would not by itself isolate the reserve-demand channel (Dilts Stedman, 2025).

Historical portfolio results remain conditional on the sample, proxy instruments, funding source, rebalancing and costs. A higher realized Sharpe ratio does not establish future dominance. The ETF backtest is an original descriptive calculation, not a guarantee of future performance or an identification of BRICS effects. The new quarterly COFER line chart, by contrast, uses 29 actual preserved observations; no missing observations or portfolio correlations are filled with assumed values. The Q4 composition chart uses the same revised vintage, while the older Q2 decomposition remains separate (Doshi et al., 2026; International Monetary Fund, n.d.).

Research Findings

Findings for Research Question 1

Research question: How significantly have major BRICS central banks reduced their holdings of U.S. Treasury securities and dollar reserves?

The country-level Treasury series include **official and private holders**; they are not identified central-bank portfolios. Consequently, this research question can be answered more directly for reported Treasury exposure than for each central bank's dollar reserve allocation. The evidence supports a large, heterogeneous reduction in selected countries' exposure, without demonstrating coordinated sales (cjseven007, 2026; McCallum et al., 2026).

Finding 1: The cut was large but uneven

From **January 2018 to December 2025**, China, Brazil, India and Russia together fell from **27.11% to 11.18% of all foreign-held Treasuries**. South Africa is excluded from this four-country calculation. All foreign holdings grew **49.65%**, while China fell **41.41%** and Brazil **36.50%**; India rose **23.06%**. These are economically large descriptive changes; no statistical-significance test or test of policy coordination has been performed. The December 2018 baseline used earlier gives different but consistent changes: China −39.1%, Brazil −44.3% and India +29.4% (cjseven007, 2026; U.S. Department of the Treasury, 2026a).

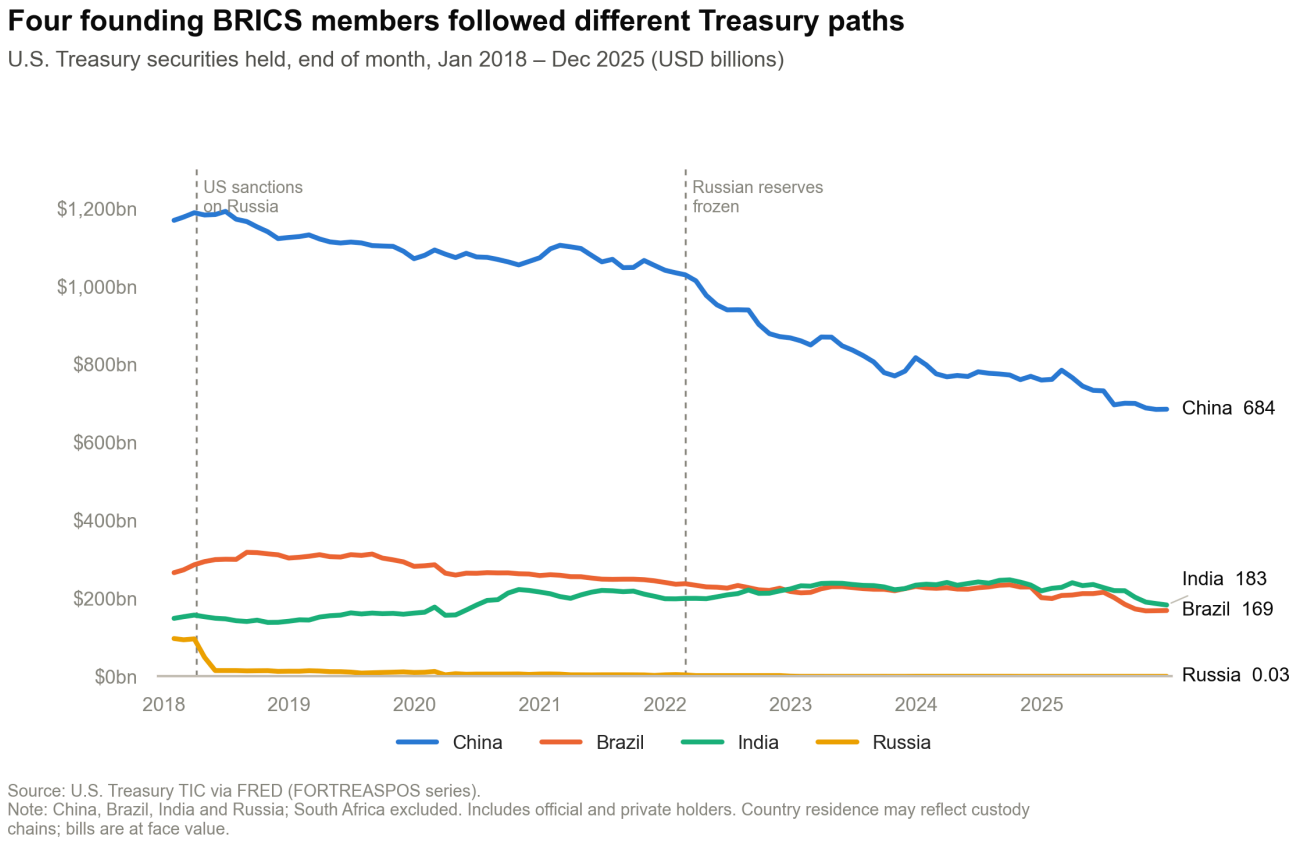


Figure 11. Four founding BRICS members are shown; South Africa is excluded. These country-attributed positions include official and private investors. Sanctions markers identify chronology, not estimated effects (cjseven007, 2026; U.S. Department of the Treasury, 2026a).

India's January-2018 index reached **166.36** in **September 2024**, when holdings were US$247.234 billion, before falling to an index of **123.06** at end-2025. China ended at **58.59** and Brazil at **63.50**. Russia's sharp initial reduction was concentrated in April–May 2018: reported holdings fell from US$96.050 billion in March to US$48.724 billion in April and US$14.905 billion in May. That was not a complete exit in a few weeks: Russia still held US$13.216 billion at end-2018 and US$0.029 billion at end-2025 (U.S. Department of the Treasury, 2026a).

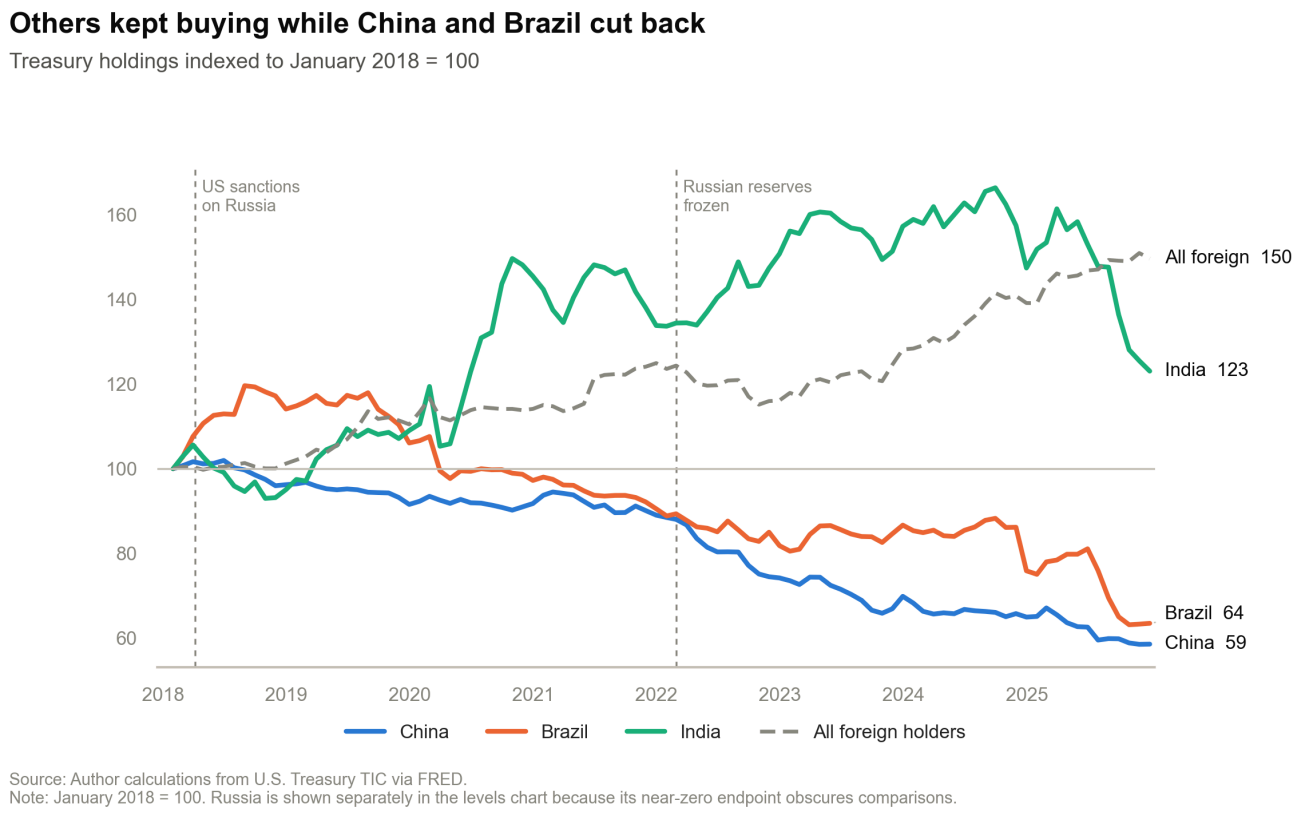


Figure 12. January 2018 = 100, which differs from the earlier December-to-December comparison. Russia is covered in the levels figure rather than this index comparison (cjseven007, 2026).

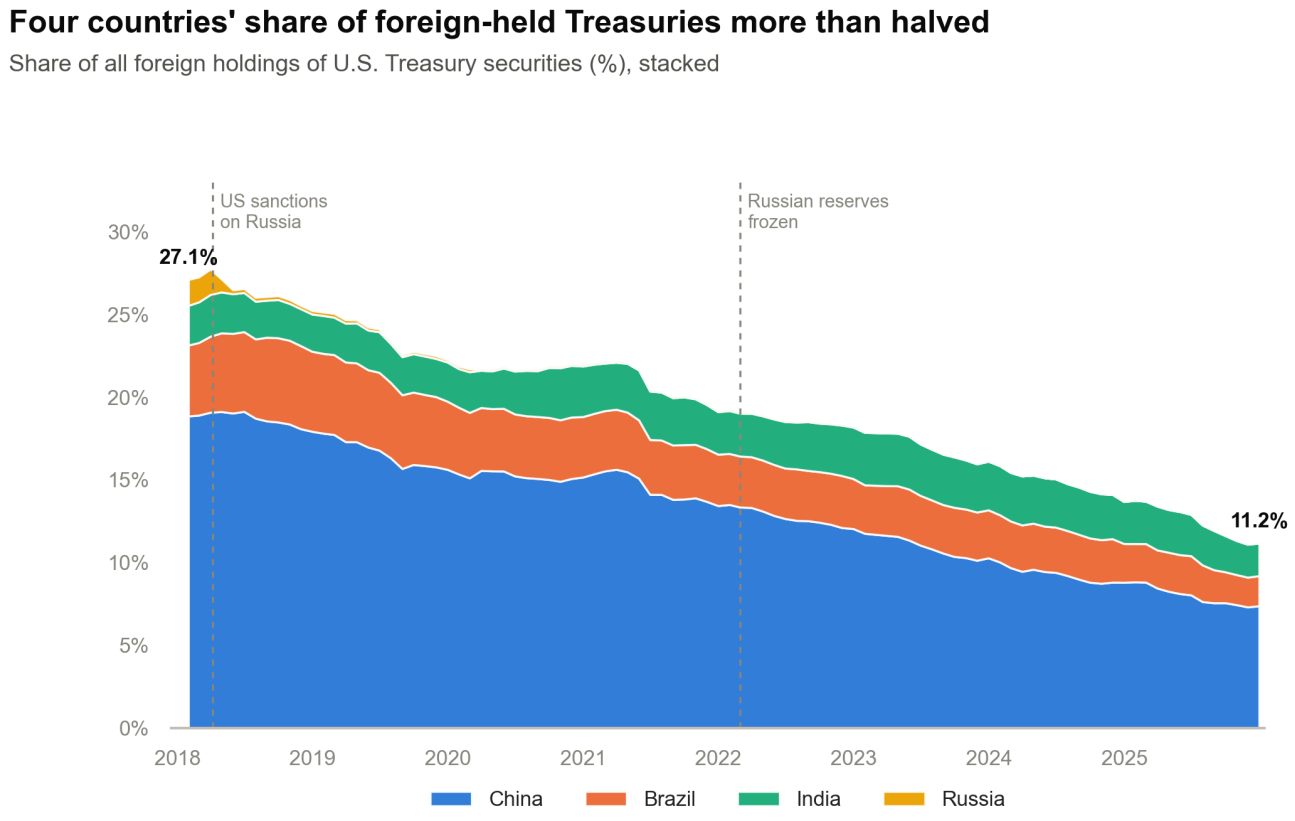


Figure 13. The denominator is all foreign Treasury holdings, not all U.S. debt, central-bank reserves or expanded BRICS membership (cjseven007, 2026).

The answer is therefore **"large but not demonstrably coordinated."** The countries followed different paths, and similar outcomes alone would not establish a common policy decision. India's increase over the full sample is particularly inconsistent with treating BRICS as one reserve manager.

Finding 2: Net sales explain most of China's long-term decline, with a substantial 2022 valuation effect

Over **calendar 2018–2025**, China's long-term Treasury holdings fell by **US$555.839 billion**, measured from the December 2017 opening position. Cumulative net transactions were **−US$533.557 billion**, valuation changes **−US$40.269 billion**, and other changes **+US$17.987 billion**:

$$\Delta H = F + \text{valuation change} + \text{other changes}$$

Here ΔH is the change in China's long-term holdings and F means net transactions. Valuation changes capture gains or losses from changing security prices; other changes retain the remaining recorded adjustments.

In US$ billions:

$$-533.557 - 40.269 + 17.987 = -555.839$$

Thus, **net sales were smaller than the long-term holdings decline**, They were larger than the **US$500.530 billion decline in all-maturity holdings**, because the implied short-term position increased by **US$55.309 billion**. This is an accounting reconciliation, not proof that each sale financed a

bill purchase. The residual must remain visible rather than being assigned arbitrarily to custodians or gold (McCallum et al., 2026; U.S. Department of the Treasury, 2026a).

Calendar year	Net transactions (US$ bn)	Valuation change (US$ bn)	Other changes (US$ bn)	Long-term holdings change (US$ bn)
2018	-51.084	-12.612	+0.000	-63.696
2019	-87.351	+37.802	+0.000	-49.549
2020	-81.892	+47.029	-0.000	-34.863
2021	+37.391	-33.869	-0.031	+3.491
2022	-59.441	-114.386	+0.000	-173.827
2023	-79.272	+11.325	+5.278	-62.669
2024	-101.791	-7.624	+8.421	-100.994
2025	-110.117	+32.066	+4.319	-73.732

Author calculations from preserved monthly CSLT series. Net transactions are purchase-positive. Long-term securities only; bills excluded. Components are rounded for display and may not sum exactly after rounding (U.S. Department of the Treasury, 2026a).

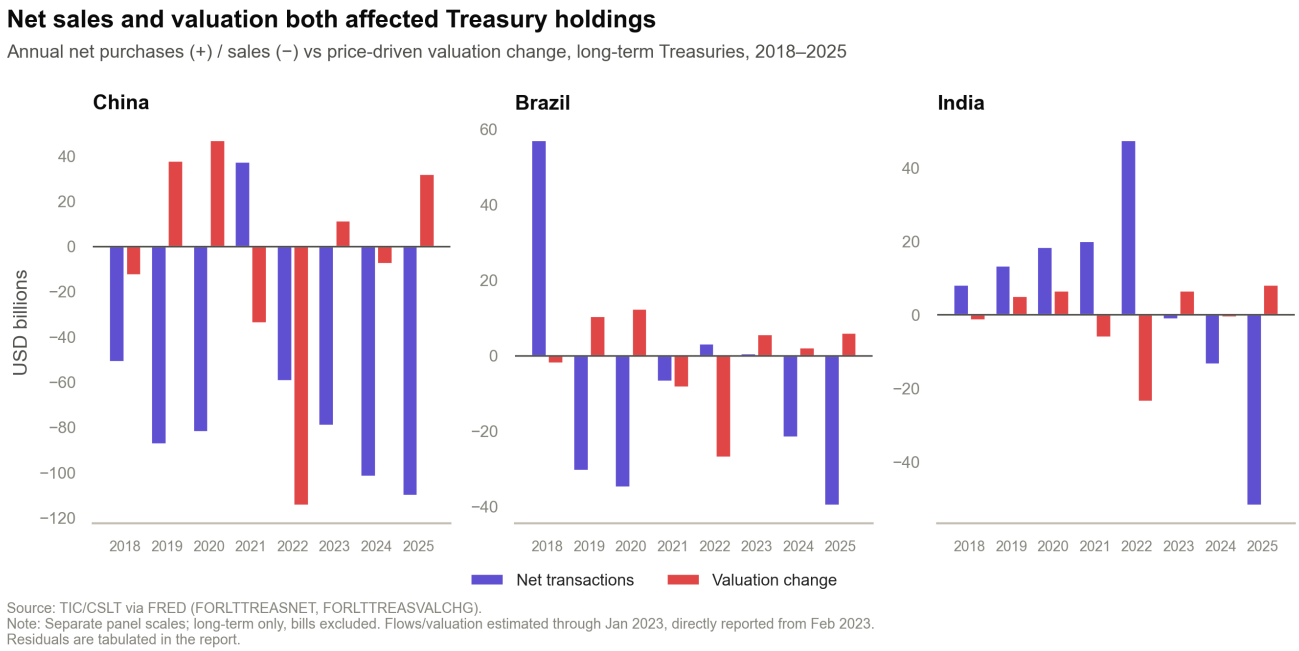


Figure 14. Purple bars are net transactions and red bars are valuation changes; each country panel has its own scale. The table retains other changes omitted from the two-bar comparison (cjseven007, 2026; McCallum et al., 2026).

China's annual net transactions were negative in every year except **2021**, ranging from roughly US$51 billion to US$110 billion of annual net sales. In **2022**, the US$114.386 billion valuation loss exceeded the US$59.441 billion of net sales, accounting for about **65.8%** of that year's long-term holdings decline. Brazil's 2022 result is more distinct: it made **US$3.197 billion of net purchases**, while a **US$26.974**

billion valuation loss produced a US$23.777 billion holdings decline. Brazil was not a net seller that year. India made **US$47.600 billion of net purchases in 2022**, then **US$51.803 billion of net sales in 2025** (U.S. Department of the Treasury, 2026a).

Measurement qualification. Through January 2023, CSLT transactions and valuation components are estimates derived from holdings, price indexes and other adjustments. Since February 2023, the expanded SLT directly collects transactions and valuation changes, with other changes calculated residually. Therefore, the pre-2023 comparison is a model-based decomposition, not an independent audit of actual trading tickets. The series also include private investors. India's dollar liquidity needs, exchange-market intervention, portfolio choices and custody changes are possible explanations for its different path, but these country-level data do not identify the RBI's motive or prove that Treasury sales funded rupee intervention (McCallum et al., 2026).

Finding 3: Gold accumulation occurred in parallel, at different times

Between **end-2018 and end-2025**, reported gold holdings increased by **453.79 tonnes in China**, **279.90 tonnes in India**, **213.49 tonnes in Russia** and **105.08 tonnes in Brazil**. South Africa changed by only **0.19 tonnes**. These changes measure reported quantities but do not identify the source of financing (cjseven007, 2026; World Gold Council, n.d.).

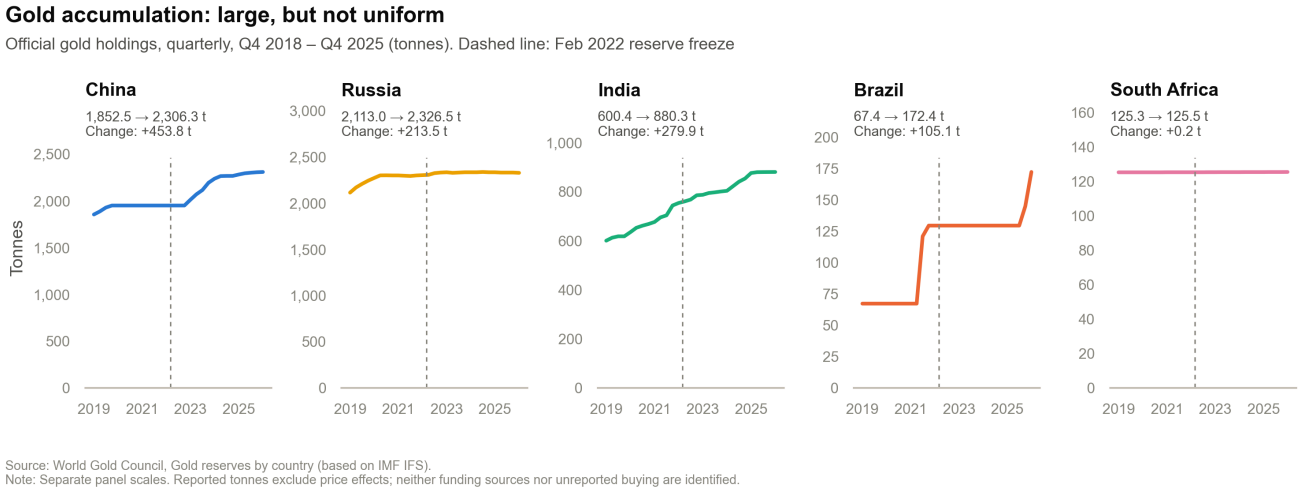


Figure 15. All five founding members are included, using 29 quarterly observations per country and separate panel scales. The February 2022 marker is contextual and does not identify a causal change (cjseven007, 2026; World Gold Council, n.d.).

China added gold during 2019, was flat from **Q3 2019 through Q3 2022**, and resumed reported accumulation in Q4 2022. Russia's rapid accumulation slowed markedly around 2020, but its reported stock subsequently changed, including an increase during 2022 and a decline in 2025; "stopped buying in 2020" is too absolute. Brazil's increases are concentrated in **2021 and late 2025**. The WGC full-year report dates Brazil's 2025 buying to September–November (World Gold Council, n.d., 2026a).

Scale illustration with matching dates. China's 453.79-tonne increase is measured between December 2018 and December 2025, so the aligned transaction window is **January 2019–December 2025**, when cumulative long-term net sales were **US$482.473 billion**. Valuing that gold increment at the observed end-2025 LBMA PM benchmark of **US$4,368 per troy ounce** gives:

$$V_{\text{gold}} = 453.79 \times 32{,}150.7466 \times 4{,}368$$
$$\approx \text{US\$}\,63.73 \text{ billion.}$$

Here V_{gold} is the illustrative value of the added gold. The inputs are tonnes, troy ounces per tonne and US dollars per troy ounce, respectively.

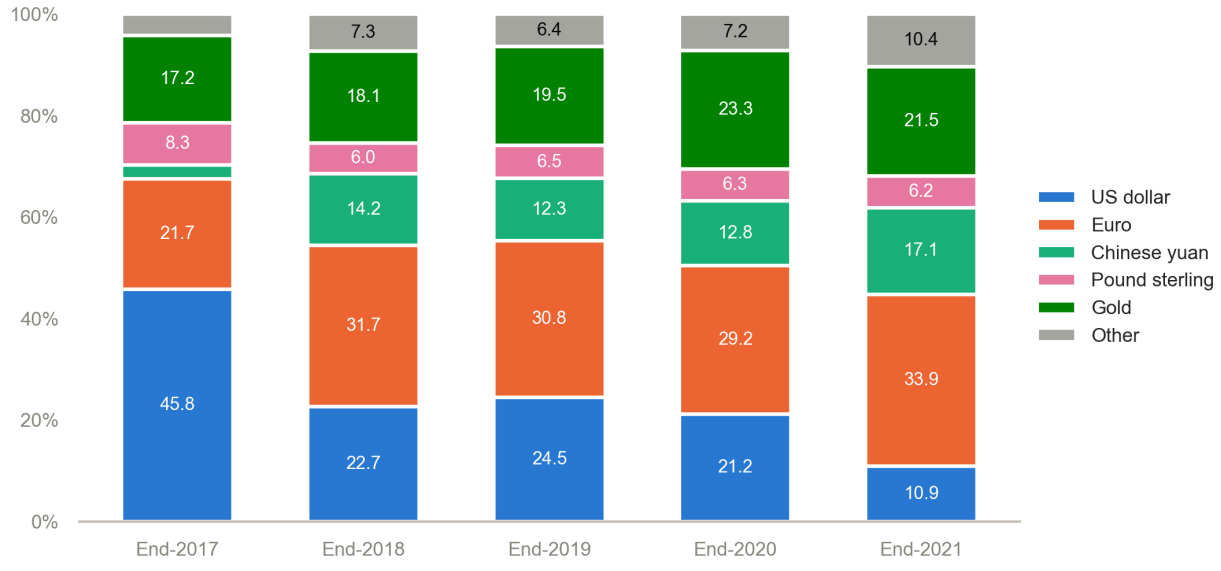
That is about **13.21%** of the aligned long-term net-sales amount. This endpoint mark-to-market comparison is **not acquisition cost, traced sale proceeds or an estimate of the fraction reinvested in gold**. Gold was acquired at different prices and times; Treasury data include private holders, whereas monetary gold is an official reserve asset. Gold therefore cannot be presented as the identified destination of the missing Treasury holdings, nor can the unexplained balance be assigned to any particular asset (U.S. Department of the Treasury, 2026a; World Gold Council, 2026b).

Finding 4: Russia demonstrates the difference between currency and custody risk

Bank of Russia disclosures show the dollar share of its foreign-currency and gold assets falling from **45.8% at end-2017** to **22.7% at end-2018** and **10.9% at end-2021**. The euro share rose from 21.7% to **33.9%** over the end-2017–end-2021 comparison. These shares include gold and unsettled FX transactions, so their denominator differs from COFER. The end-2019 disclosure adds a dollar share of 24.5%, euro 30.8%, gold 19.5%, renminbi 12.3%, sterling 6.5%, and other currencies 6.4%. The disclosure dates are 1 January of the following year, interpreted here as the preceding year-end (Bank of Russia, 2019, 2020, 2022).

Russia cut its dollar share well before the 2022 freeze

Bank of Russia foreign currency and gold assets by currency (% of market value)



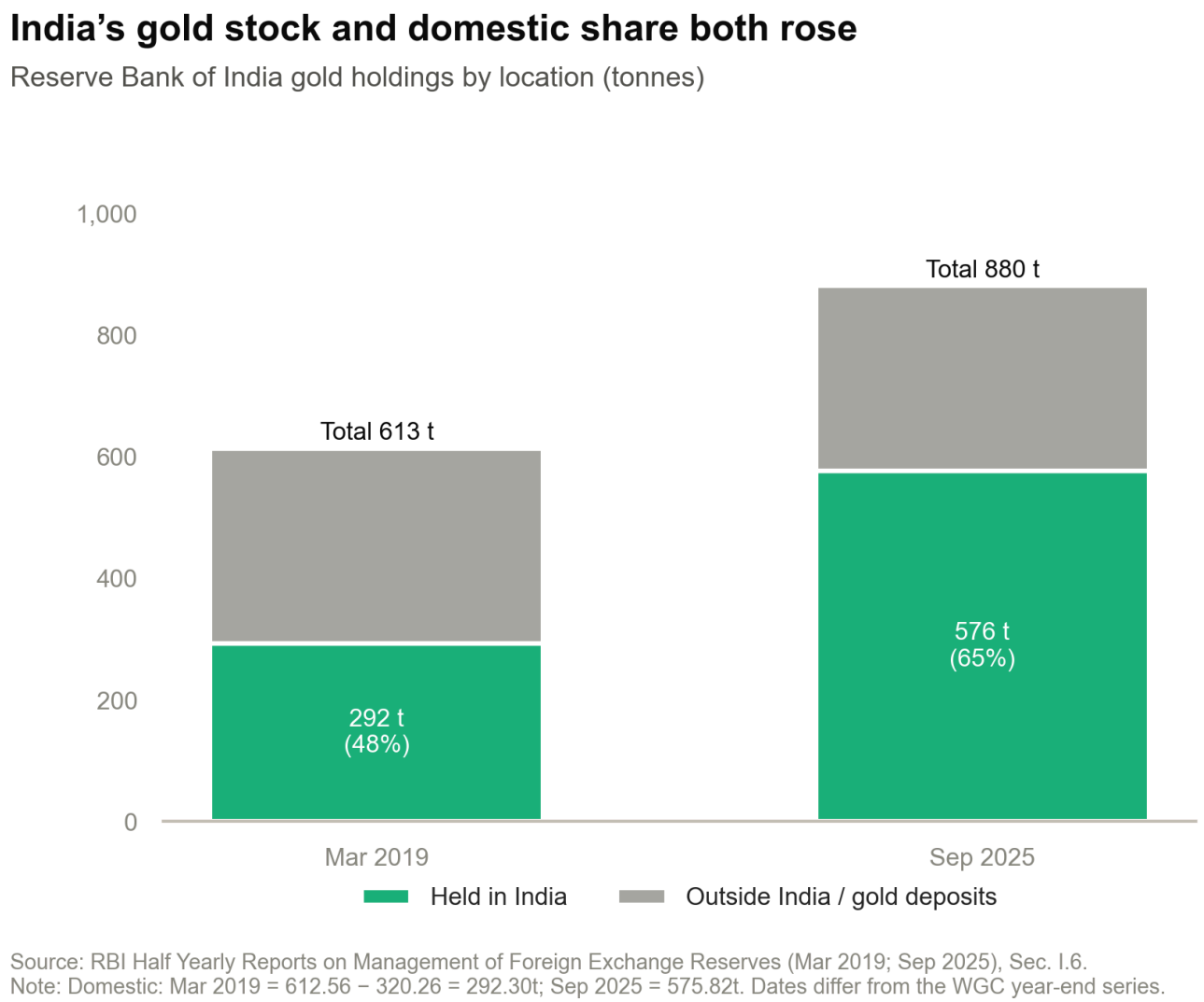
Source: Bank of Russia Annual Reports: 2018 Chart 32, 2019 Chart 37, 2021 Chart 32.
Note: Gold included, unlike COFER. Year-end observations shown; no interpolation. Values below 5% not printed.

Figure 16. Source: Bank of Russia annual reports for 2018 (Chart 32, p. 86), 2019 (Chart 37, printed p. 99; PDF p. 100) and 2021 (Chart 32, printed p. 102; PDF p. 104). End-2017 is the opening baseline. Currency denomination does not identify custody location (Bank of Russia, 2019, 2020, 2022).

The subsequent restrictions on transactions with the Bank of Russia show why changing a reserve asset's denomination is insufficient to remove jurisdictional risk: euro-denominated assets can also be immobilized through institutions in sanctioning jurisdictions. Holding physical gold domestically reduces

exposure to a foreign custodian's immobilization, but **does not make gold immune to sanctions, settlement constraints or difficulty realizing liquidity**. OFAC's published guidance explicitly includes gold-related transactions in restrictions involving the Bank of Russia (Office of Foreign Assets Control, 2022).

India provides a separate custody observation. Domestic gold rose from a derived **292.30 tonnes out of 612.56 tonnes at 31 March 2019 (47.72%)** to **575.82 out of 880.18 tonnes at 30 September 2025 (65.42%)**. For 2019, domestic gold equals total holdings minus 320.26 tonnes held overseas. The September 2025 balance outside domestic custody comprises 290.37 tonnes held with the BoE/BIS and 13.99 tonnes of gold deposits. These observations demonstrate a shift toward domestic storage; they do not identify how much was repatriated rather than bought and stored domestically, or establish a common motive with China or Russia (Reserve Bank of India, 2019, 2025).



Source: RBI Half Yearly Reports on Management of Foreign Exchange Reserves (Mar 2019; Sep 2025), Sec. I.6.
Note: Domestic: Mar 2019 = 612.56 − 320.26 = 292.30t; Sep 2025 = 575.82t. Dates differ from the WGC year-end series.

Figure 17. Exact observation dates differ from the WGC year-end quantity series. Domestic and other holdings sum to the reported total; rounded chart labels do not replace the precise values in the text (Reserve Bank of India, 2019, 2025).

Finding 5: Globally, the dollar slipped, while the yuan's share was approximately unchanged

In the **30 September 2026 COFER release**, the dollar share fell from **62.07% in 2018 Q4 to 56.52% in 2025 Q4**, a **5.54-percentage-point decline calculated before rounding**. The renminbi share was approximately **1.96% at both endpoints**. "Other currencies" gained **3.55 percentage**

points, the largest offsetting increase; yen, Canadian dollar and Australian dollar shares also rose. This is consistent with diversification into a wider currency set, including the nontraditional currencies discussed by Arslanalp et al. (2022). It does not show that individual dollars sold were transferred to a particular currency or that the yuan replaced the dollar globally (International Monetary Fund, 2026a).

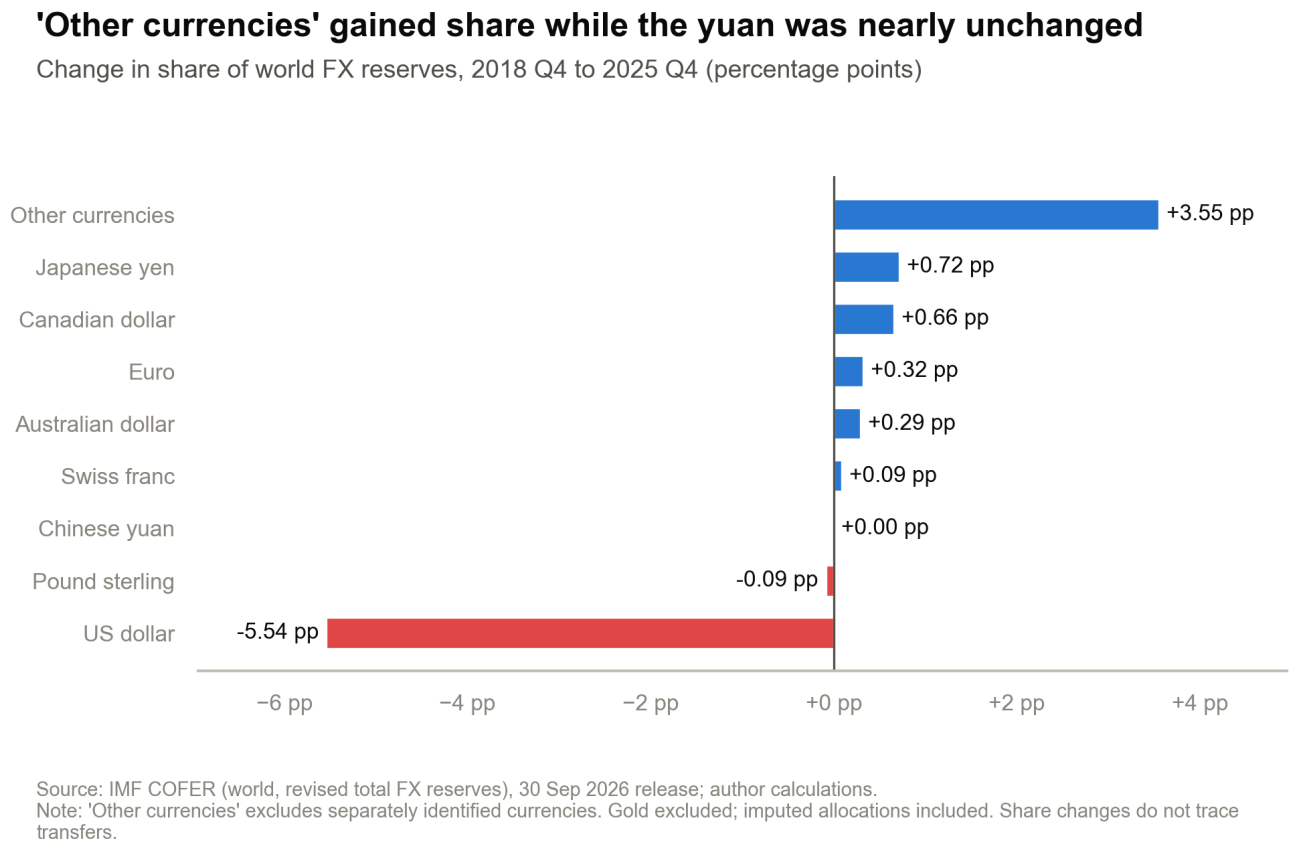


Figure 18. Changes use unrounded shares from one consistent release; the nine currency categories sum to 100% in each quarter. “Other currencies” is separate from individually identified AUD, CAD and other currencies (cjseven007, 2026; International Monetary Fund, 2026a).

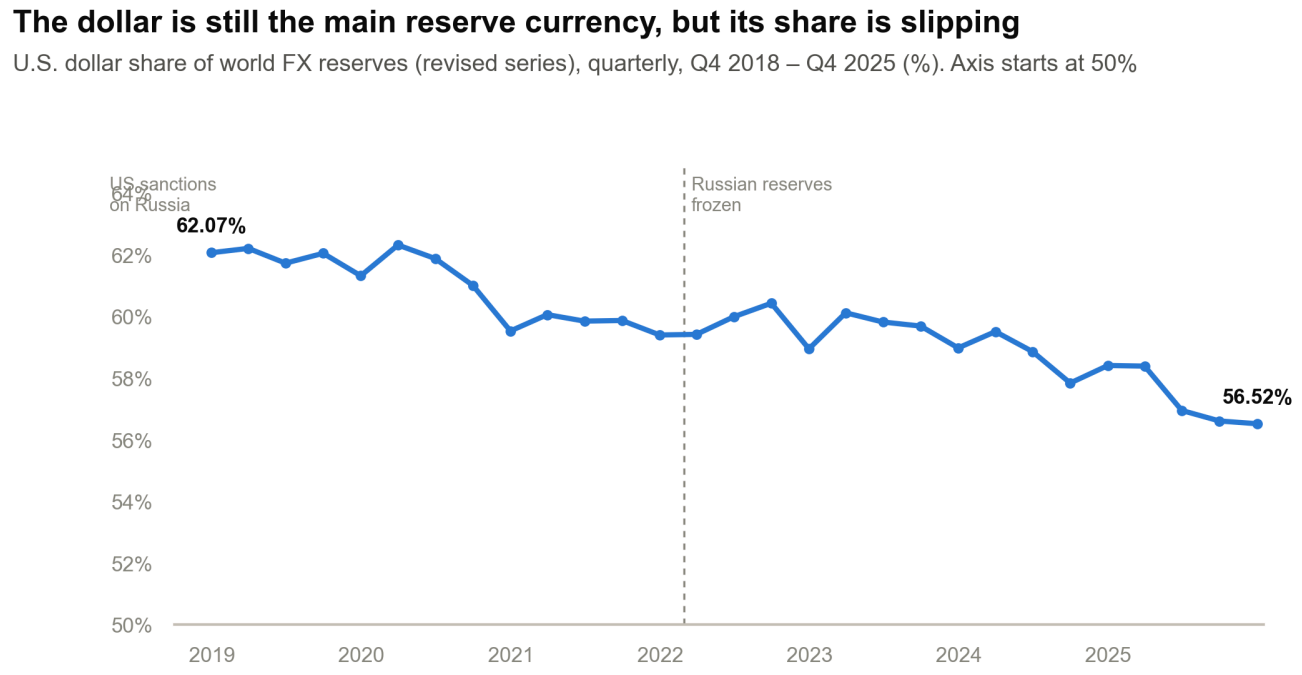


Figure 19. The axis begins at 50%, as explicitly labelled. The revised series includes imputed currency allocations and excludes gold. It is not spliced with the older October 2025 allocated-reserves snapshot (International Monetary Fund, 2026a).

Two qualifications prevent a sales interpretation. First, dollar-denominated reserve value actually **rose from approximately US$7.089 trillion to US$7.431 trillion** in this release, despite its lower share of a growing total. Second, exchange-rate valuation affects the shares; the separate historical Q2 decomposition shows how large that effect can be. The 56.52% value does not disclose the currency allocation of an individual BRICS central bank (International Monetary Fund, 2026a, 2026b; Kwende et al., 2025).

Finding for RQ1. The defensible conclusion is:

> Major BRICS economies have not followed a uniform de-dollarization path. China and Brazil materially reduced their directly reported Treasury exposure, Russia's position became negligible after a particularly sharp 2018 reduction, while India ended above its January and December 2018 baselines. The CSLT decomposition supports a dominant net-sales contribution to China's long-term decline, subject to pre-2023 estimation and country-attribution limits. Reported gold accumulation and a broader global reserve-currency mix support diversification, but neither identifies Treasury-sale proceeds, synchronized BRICS policy, a yuan replacement or a measured central-bank dollar-allocation change for every member.

The Federal Reserve's 2025 review also finds continued dollar leadership across international reserves and transactions. Its evidence largely predates the 2025 policy changes discussed in that note, so it is a baseline rather than a test of their eventual effects (Bertaut et al., 2025).

Findings for Research Question 2

Research question: To what extent has the reduction in foreign sovereign demand influenced benchmark 10-year U.S. Treasury yields and bond prices?

Country and aggregate evidence. China's large cumulative net sales are compatible with the Kansas City Fed's aggregate official-investor valuation result. RQ1 covers mainland-China-attributed **official and private holders**, long-term securities, and calendar 2018–2025; the published chart covers an **aggregate foreign-official** population and its own source period. Other countries and private buyers can offset China's sales. In 2022, valuation losses also explain more of China's long-term decline than its net sales, while Brazil bought on net. Neither the country transactions nor the aggregate chart identifies the BRICS central-bank component of a demand shock (Dilts Stedman, 2025; McCallum et al., 2026).

The decomposition distinguishes holdings changes from transactions, but it does not establish exogeneity, official ownership or a yield coefficient. Estimated pre-2023 flows and the measurement transition would require explicit treatment in any future regression. In particular, dividing multi-year Chinese sales by the literature's one-month US$141 billion shock and multiplying by 57 basis points would still be invalid. The published yield range and duration calculations below remain conditional scenarios.

The economic mechanism is theoretically sound: when a price-insensitive buyer reduces demand, a larger share of the available Treasury supply must be absorbed by investors whose demand depends more strongly on expected return. All else equal, the equilibrium price is lower and the yield higher.

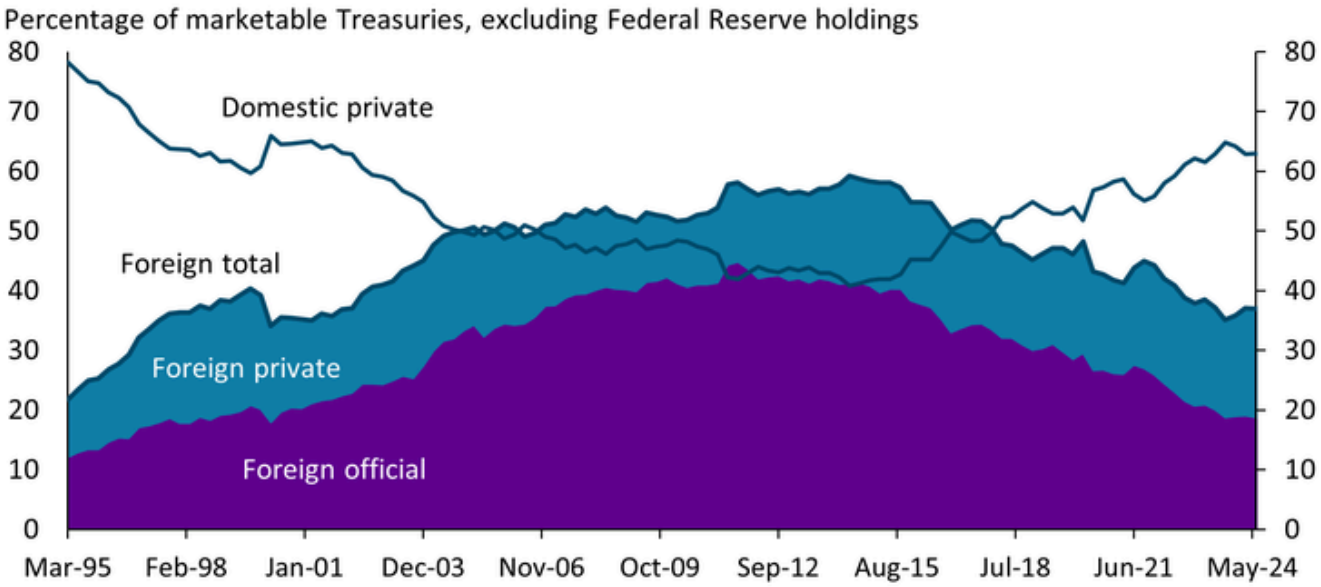


Figure 20. Source: (Dilts Stedman, 2025). Its denominator is marketable Treasuries excluding Federal Reserve holdings, which differs from the all-foreign-holdings denominator in Figure 2. Historical context extends beyond this report's 2018–2025 country sample.

Published studies support the existence of this effect, but not one universal coefficient. Warnock and Warnock's historical work found economically large yield effects from foreign capital flows, and later research has used alternative flow definitions and identification strategies. The 2025 Kansas City Fed comparison shows that estimates for a standardized foreign Treasury sale vary widely; its cross-study average is around 57 basis points for a one-standard-deviation liquidation of roughly US$141 billion, with an estimated range of about 25–101 basis points. Dilts Stedman (2025, p. 2) explicitly presents this as a synthesis across studies, not a BRICS-specific causal estimate (Dilts Stedman, 2025).

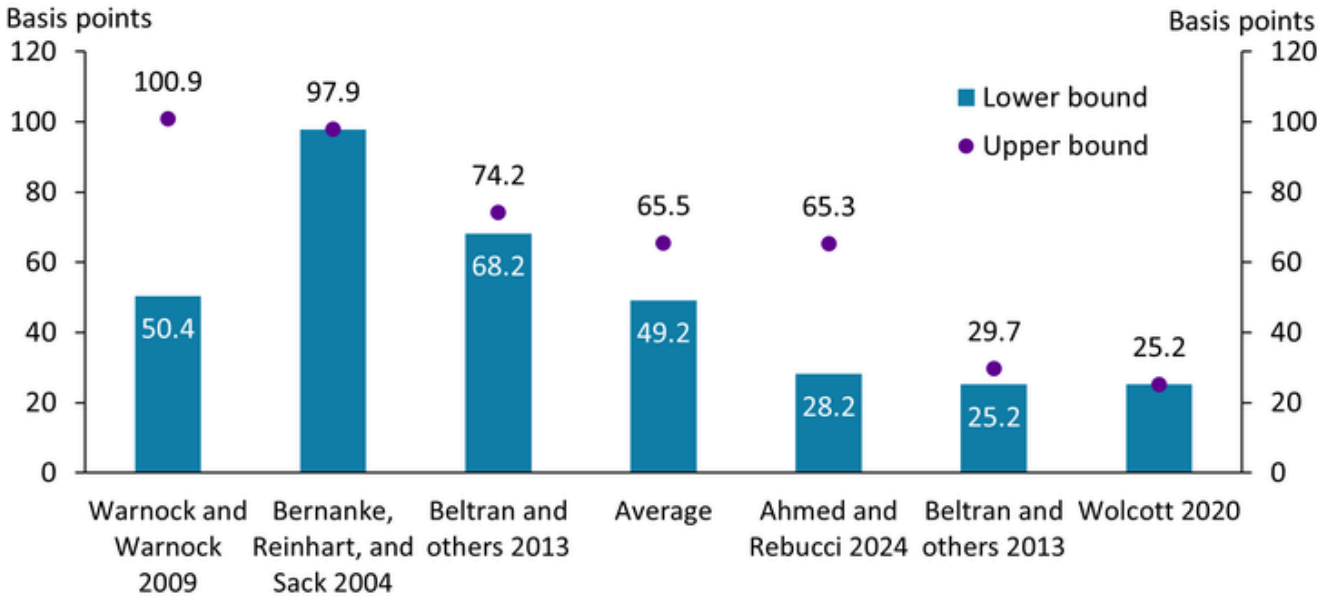


Figure 21. Source: (Dilts Stedman, 2025). Estimates standardize a one-standard-deviation foreign Treasury sale, approximately US$141bn as of March 2025. Study maturities and flow/holdings definitions differ; these are literature estimates rather than measured BRICS effects.

This range itself is analytically important. Suppose a report simply takes the 57-basis-point average and multiplies it by an observed decline in China's holdings. That procedure would be invalid for several reasons:

- the decline in holdings may not equal a transaction flow;
- the empirical studies use different maturities, aggregation methods and identification assumptions;
- a multi-year change is not equivalent to a one-month liquidation shock;
- private investors can offset foreign-official selling;
- the Treasury stock and issuance volume change over time; and
- yields simultaneously respond to Federal Reserve policy, inflation, fiscal supply and growth expectations.

The Kansas City Fed decomposition makes the problem particularly concrete. Its analysis shows that aggregate foreign official Treasury allocations have been comparatively stable since around 2013 and that much of the decline in the **value** of official Treasury holdings since 2020 reflected passive valuation effects. It also notes that the large expansion of Treasury issuance during the pandemic was not matched by a proportional increase in official foreign demand (Dilts Stedman, 2025).

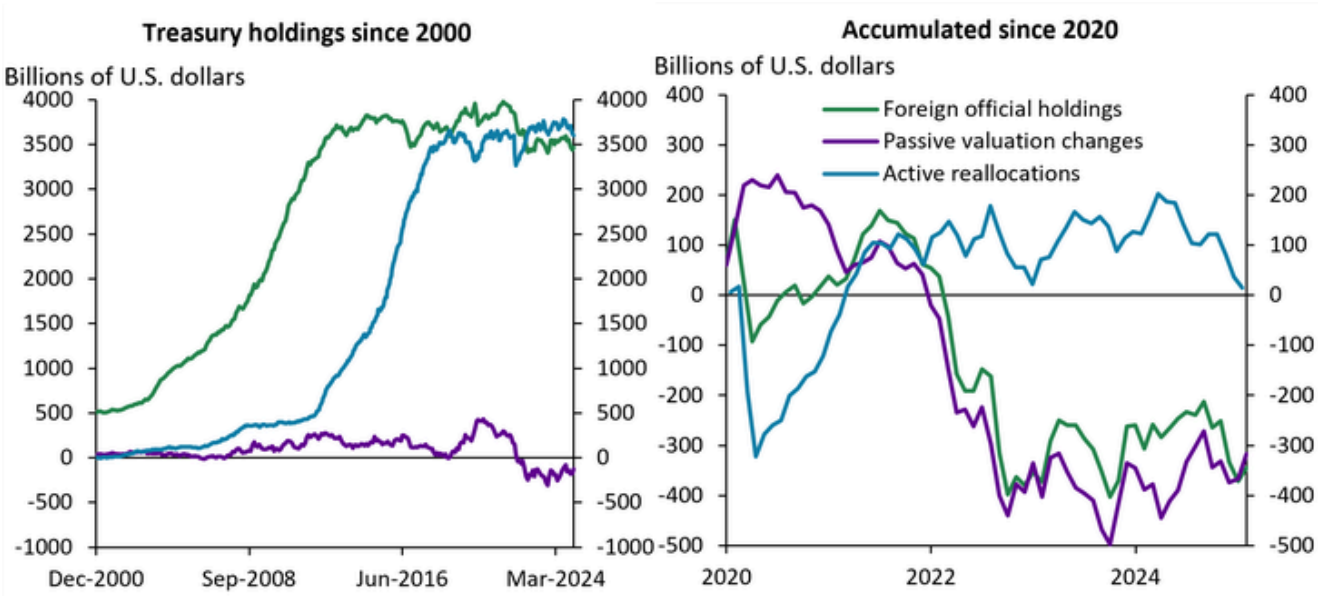


Figure 22. Source: (Dilts Stedman, 2025). The blue active-reallocation series differs from the green holdings series and purple valuation series. This aggregate foreign-official analysis is not a decomposition of China's individual position.

This means "absence of additional buying" can matter even without net selling. Consider two equilibria:

Case 1: Treasury supply rises, and official purchases rise by the same proportion.

versus

Case 2: Treasury supply rises, while official purchases stay unchanged.

The second case forces other investors to absorb the marginal supply. Therefore, the economically relevant concept is **net demand relative to supply**, not simply whether a country's holdings are falling.

Modeled scenario. Using a 7.5 modified duration, the Kansas City Fed's 57-basis-point cross-study average would correspond to an approximate first-order bond-price effect of:

$$\text{Price change} \approx -7.5 \times \frac{57}{100} = -4.275\%$$

This **does not mean BRICS caused a 4.28% Treasury loss**. It means that *if* an exogenous demand shock caused a parallel 57-basis-point yield increase, a bond with 7.5 modified duration would experience approximately that first-order price decline before convexity and income.

Sensitivity to different yield effects

Hypothetical demand-induced yield shock	Approximate loss, duration = 6.5	duration = 7.5	duration = 8.5
+25 bp	−1.63%	−1.88%	−2.13%
+50 bp	−3.25%	−3.75%	−4.25%
+57 bp	−3.71%	−4.28%	−4.85%
+75 bp	−4.88%	−5.63%	−6.38%
+100 bp	−6.50%	−7.50%	−8.50%

The table demonstrates a higher-order investment implication: **de-dollarization risk is not the same for all Treasury investors**. A short-duration investor can face little mark-to-market sensitivity even when a long-duration investor experiences substantial price volatility.

A rise in the 10-year Treasury yield does not "mechanically" reduce all equity valuations one-for-one. A higher risk-free component can increase discount rates, but simultaneously changing expected cash flows, inflation, term premiums and equity risk premiums can reinforce or offset the effect. CAPM is useful for conceptualizing required return:

$$\mathrm{RRR} = \mathrm{RF} + \beta \times \mathrm{RP}$$

Here RRR is the required return, RF is the risk-free rate represented by Treasury bills, and β measures sensitivity to the market. RP is the expected market risk premium: expected market return minus the risk-free rate.

A β of 1 means the same market sensitivity as the market itself in this model. **Investors require compensation for taking market risk, but the equation does not predict what caused that risk to change.** Use the same time period for all returns.

CAPM does not establish that foreign Treasury sales cause a predictable percentage fall in equity prices.

Who absorbs Treasury supply? A reduction in official buying does not mechanically determine a yield change. If other buyers absorb the additional supply at prevailing prices, a decline in official demand need not raise yields; otherwise the required compensation and yield may rise. Dealer capacity, investor risk appetite, maturity and market conditions matter. A sustained increase in supply can require more absorption by other investors even if official investors do not sell. Figure 38 presents this as a conditional market-clearing mechanism, not an estimated model (Dilts Stedman, 2025).

TIC/CSLT data show approximately **US$78.9 billion of cumulative foreign-official net purchases of long-term securities** during 2018–2025. Annual transactions include –US$121.8 billion in 2020 and +US$183.1 billion in 2021. "Other foreign" in Figure 28 is total foreign less official; it includes foreign private investors, not U.S. investors. Before February 2023, the transaction observations are estimated; later CSLT observations are collected directly. Figure 29 compares foreign positions with the broader marketable Treasury liability series only as context: different coverage, valuation and maturity make subtraction inappropriate (McCallum et al., 2026; Federal Reserve Bank of St. Louis, n.d.).

Yield components and maturity. A useful conceptual decomposition is:

$$\text{Long-term yield} \approx \text{expected short-term rates} + \text{term premium}.$$

Both components in Figure 30 are Kim–Wright model estimates, rather than directly observed quantities; the fitted zero-coupon yield is not the same series as the observed constant-maturity par yield shown separately in Figure 39. In the month-end model data, the fitted ten-year yield rose from 3.83% in June 2023 to 4.95% in October, while its estimated term premium rose from 0.10% to 0.62%. These monthly observations do not identify a BRICS effect. For the different July 26–October 19 window, Diercks and Asnani (2024) discuss an estimated term-premium contribution of about 0.69 percentage points and an expectations contribution of about 0.44 points under this model; other models differ.

Beltran and Li (2026) estimate that a US$100 billion **supply shock** increases the five-year yield by roughly 3 basis points in their model. The Kansas City Fed synthesis instead standardizes a roughly US$141 billion one-month **foreign sale**, across studies and maturities, with an average around 57 basis points. These are different shocks, periods, maturities and methodologies; neither is a coefficient to apply mechanically to BRICS holdings.

New sources of dollar demand. Ahmed and Aldasoro's (2026) June 2026 revision estimates that a US$3.5 billion stablecoin inflow lowers three-month Treasury bill yields by about 0.71 basis points immediately and up to 4 basis points within ten days. The paper reports roughly US$153 billion in bill holdings in late 2025 and US$33 billion of purchases over one year (Figure 31). Its USDT and USDC observations end in December and November respectively and include indirect bill exposure. The two amounts are a stock and a change in positions; the paper studies bill markets, not ten-year notes or BRICS sales. Weiss (2026) also reports nearly US$200 billion of foreign-official purchases of Treasuries across **all maturities** from 2022 through April 2026: that wider window and coverage cannot be equated to the long-term 2018–2025 series here.

Finding for RQ2.

> Foreign official Treasury demand is economically capable of influencing U.S. yields, and published research indicates that sufficiently large exogenous foreign sales can raise medium- and long-term yields. However, the 2018–2025 evidence does not support attributing a precise fraction of observed 10-year yield movements to BRICS reserve changes. The appropriate investment conclusion is therefore a conditional one: reduced structural official demand can increase the yield required from marginal private buyers, creating additional duration risk, but Federal Reserve policy, inflation, fiscal issuance and changing term premiums prevent a clean BRICS-specific causal estimate.

Findings for Research Question 3

Research question: How does the shift away from U.S. dollar dominance impact the risk, return and diversification benefits of a multi-asset investment portfolio?

Gold custody and portfolio exposure. Reported official gold accumulation documents a reserve-management choice; it does not measure the effect of de-dollarization on private portfolio returns. India's domestic storage share and Russia's jurisdictional exposure concern sovereign custody arrangements. The GLD exposure in the published simulation is an exchange-traded vehicle with its own custody, legal, fee and liquidity arrangements; it is not equivalent to a central bank holding bullion in domestic vaults. Consequently, RQ1 does not establish that GLD inherits sovereign physical-gold custody protection, or that Treasury sales caused gold's historical performance. The ETF backtest and correlation matrix below measure portfolio outcomes rather than sovereign custody effects (Doshi et al., 2026; Reserve Bank of India, 2019, 2025).

The relevant concept is **covariance**, not the geopolitical label attached to an asset.

The earlier portfolio-risk equation has a straightforward implication: the result depends on the weight of each asset, its volatility, and the correlation between each pair. Correlation runs from -1 to $+1$. A value near $+1$ means returns tend to move together; a value near -1 means they tend to move in opposite directions.

Adding gold helps when its behaviour complements the assets already held. A low correlation can be useful even when gold itself is volatile; changing the funding source changes the portfolio being compared.

A gold allocation can reduce portfolio risk even if gold itself is volatile, provided its covariance with the existing portfolio is sufficiently low. Conversely, an asset called a “safe haven” can still fail to improve a portfolio if purchased at too high a weight or if its correlations change.

Peer-reviewed work by Baur and Lucey (2010, pp. 217–229) supports a conditional characterization: gold can act as a hedge and, during certain extreme equity-market episodes, a safe haven, but the safe-haven property is not permanent. That is considerably more defensible than describing gold as universally superior to sovereign bonds (Baur & Lucey, 2010).

A useful published benchmark validates the portfolio mechanism. State Street's December 2004–December 2025 monthly backtest used the S&P 500 Total Return Index, Bloomberg U.S. Aggregate Bond Total Return Index and GLD, with annual rebalancing. Its results are:

Portfolio implementation	CAGR	Annualized volatility	Sharpe ratio	Maximum drawdown
60/40, no gold	8.05%	9.34%	0.64	−30.75%
57.5/37.5/5, even-funded gold	8.32%	9.08%	0.69	−28.92%
55/35/10, even-funded gold	8.58%	8.90%	0.73	−27.06%
60/35/5, gold funded from bonds	8.50%	9.40%	0.68	−30.30%
60/30/10, gold funded from bonds	8.95%	9.53%	0.72	−29.85%

Published historical simulation, not an original 2018–2025 backtest. It assumes monthly data, annual rebalancing, and no transaction costs or taxes. (Doshi et al., 2026).

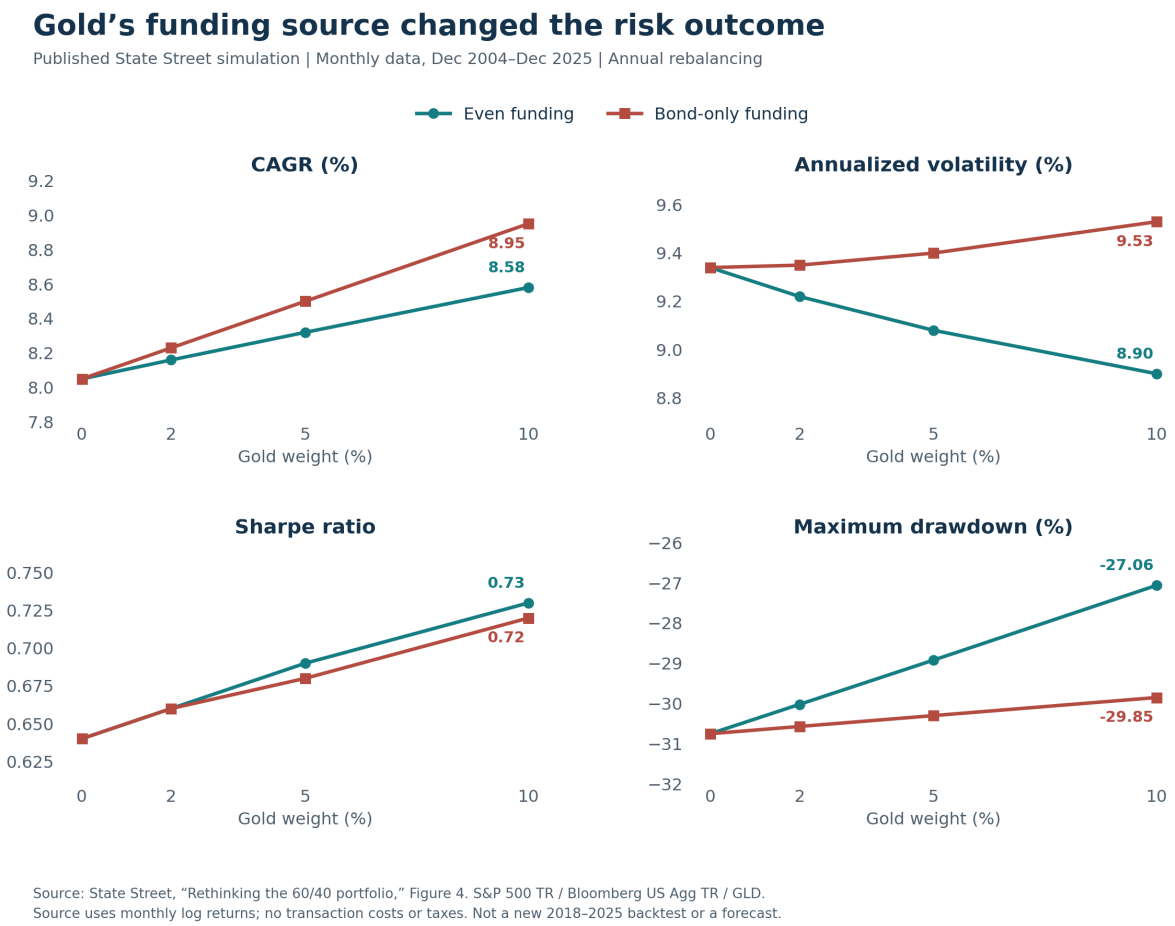


Figure 23. Author chart of (Doshi et al., 2026), including its published 2%, 5% and 10% GLD allocations. Panels show CAGR, annualized volatility, Sharpe ratio and maximum drawdown. Source simulation uses monthly log returns from December 31, 2004 to December 31, 2025 and annual rebalancing; this is not a newly run 2018–2025 test.

This table produces a more interesting result than “gold is good.”

When gold is funded **evenly** from stocks and bonds, historical volatility declines as the gold allocation rises. When gold is funded **only from bonds**, return rises in the historical sample, but volatility also rises. In other words:

A larger gold allocation does not guarantee lower portfolio volatility.

The funding source is part of the investment decision.

The drawdown evidence is similarly conditional. During State Street's defined 2022 Russia/inflation stress period, the month-end maximum drawdown improved from −16.1% for the zero-gold portfolio to −14.7% for a 10% even-funded gold portfolio. This supports diversification during that episode but does not prove that gold will outperform in the next inflation or geopolitical shock (Doshi et al., 2026).

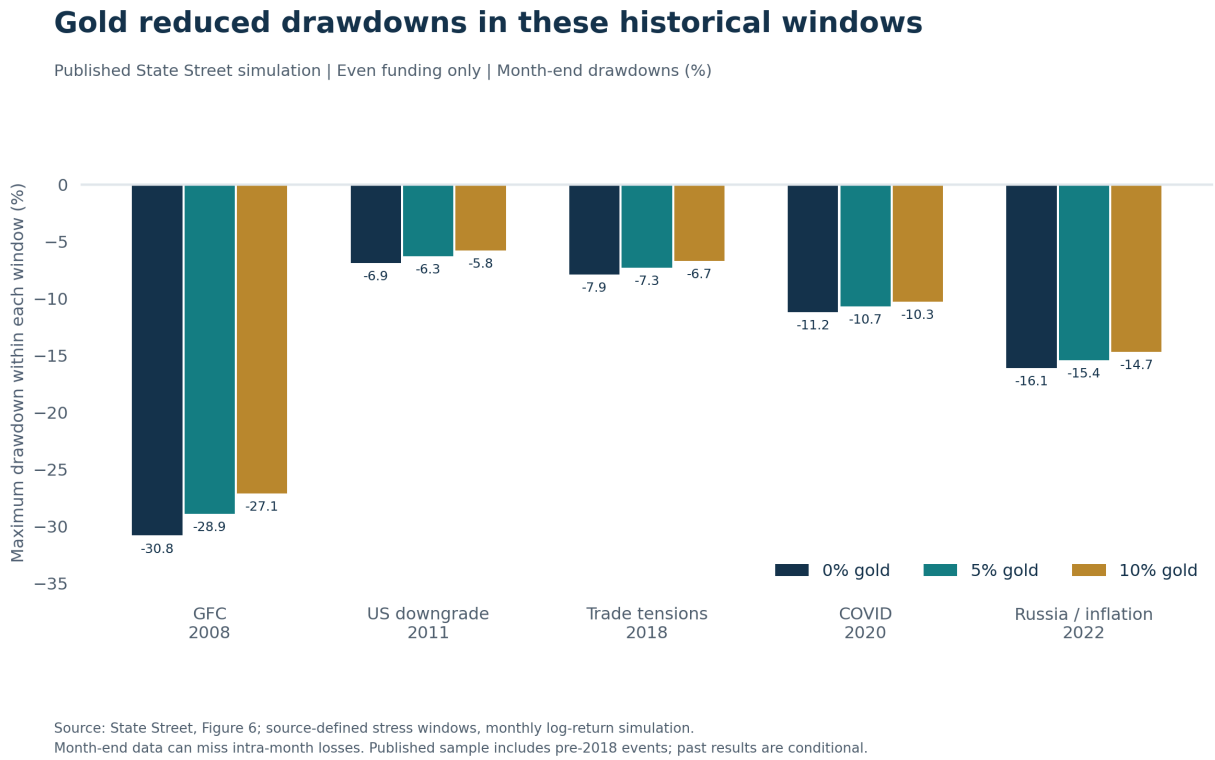


Figure 24. Author chart of the 0%, 5% and 10% even-funded columns in (Doshi et al., 2026). Stress windows are defined by the publisher; several predate the core research sample. Month-end drawdowns can miss sharper intra-month losses.

The following hypotheses motivate comparison of the calculated Sharpe ratios:

The starting claim is $S_{\text{gold}} \leq S_{60/40}$: adding gold does not improve the Sharpe ratio. The alternative is $S_{\text{gold}} > S_{60/40}$: it does improve that ratio. Here S means Sharpe ratio. The ratios below are calculated descriptively; no statistical significance test or p-value is claimed.

But Sharpe ratio alone is insufficient. A portfolio can achieve a higher Sharpe ratio while suffering a worse tail outcome or while relying on one unusually strong asset regime. The final assessment should compare Sharpe ratio, maximum drawdown, volatility and the correlation matrix together.

A regime-based interpretation is useful:

Economic regime	Treasury bonds	Gold	Expected diversification implication
Disinflation / recession	Often benefits from falling rates	Mixed	Bonds may be superior ballast
Inflation shock	Nominal bonds vulnerable	Can benefit, but not guaranteed	Gold may improve diversification
Deflationary liquidity crisis	High-quality Treasuries often strong after liquidity stabilizes	Mixed/positive	Both can diversify equities differently
Geopolitical reserve stress	Treasury response depends on flight-to-quality versus supply/risk-premium effect	Potential safe-haven demand	Gold may provide non-sovereign risk exposure

Economic regime	Treasury bonds	Gold	Expected diversification implication
Strong real growth / rising real yields	Bonds may weaken	Gold may face opportunity-cost pressure	Equities may dominate both defensive assets

The table demonstrates why a recommendation to remove Treasuries entirely would be difficult to justify. Treasuries and gold hedge **different risks**. Gold lacks contractual coupon income and has no maturity value; Treasuries have explicit dollar cash flows and deep liquidity but expose investors to duration and inflation risk. The portfolio problem is therefore complementary rather than binary.

Original-sample ETF exercise. The following calculations use SPY for stocks, AGG for aggregate bonds and GLD for gold; IEF substitutes for bonds in the sensitivity files. Adjusted monthly prices are used as total-return proxies, with annual December rebalancing, drifting weights within each year, reinvestment and no additional trading costs or taxes. ETF expenses are reflected in fund returns. BIL monthly returns provide a U.S. cash proxy for the Sharpe ratio; this is not a guaranteed risk-free return. Each case has 96 monthly observations. These results supplement, rather than replace, State Street's published 2004–2025 index-based comparison (Doshi et al., 2026; Yahoo Finance, n.d.).

Portfolio	CAGR	Annual volatility	Maximum monthly drawdown	USD Sharpe
60/40	9.38%	10.97%	−20.11%	0.654
5% gold, even funded	9.83%	10.63%	−19.63%	0.710
10% gold, even funded	10.26%	10.34%	−19.15%	0.767
5% gold, bond funded	10.13%	10.95%	−19.87%	0.719
10% gold, bond funded	10.88%	10.97%	−19.62%	0.780

The bond-funded 10% case has essentially unchanged volatility relative to 60/40 in this sample (10.9690% versus 10.9697% before rounding). In the published longer sample, its volatility increased from 9.34% to 9.53%. Differences in dates, funds versus indexes, rebalancing and return conventions, and the cash proxy matter; these results do not establish which allocation will perform better. Figures 32–33 show the original exercise.

Regimes and correlation. SPY and AGG monthly returns had a 0.422 correlation over 2018–2025, but the 2022–2023 correlation was 0.751 versus −0.222 in 2018–2019. The 24-month rolling estimates in Figure 34 illustrate changing co-movement, not a causal regime test. Li and Yoldas (2026) report correlations between **stock returns and yield changes**, a different measure: −0.26 before 2000, +0.40 in 2000–2021 and −0.09 after 2021 through July 2026 (Figure 35). Because bond prices generally move inversely with yields, a positive stock-return/yield-change correlation generally indicates stocks and bond prices moving in opposite directions. The 2026 portion is contextual and outside the core sample.

Figure 36 uses two **hypothetical**, not historical or forecast, shocks. With stocks −10%, bonds −6% and gold +8%, the 60/40 and 10% even-funded portfolios return −8.4% and −6.8%. With stocks −10%, bonds +8% and gold +2%, the same portfolios return −2.8% and −2.5%, while 10% bond-funded returns −3.4%. Results follow the stated weights and assumptions; they are not estimated effects of reserve changes.

Investor currency. A safe dollar asset is not necessarily stable in a Malaysian investor's home currency. Using unhedged MYR per USD exchange rates, the 60/40 ETF portfolio's CAGR was 9.42%, volatility 10.08%, and maximum month-end drawdown −11.59%; the 10% even-funded case was 10.30%, 9.31% and −10.22%. These outcomes include exchange-rate movements and use U.S. equity-close dates with Federal Reserve New York noon FX observations; the timing is not exactly synchronized (Figure 37). A 4% USD gain combined with an 8% dollar depreciation gives $1.04 \times 0.92 - 1 = -4.32\%$ in MYR. A USD cash proxy is not a Malaysian risk-free benchmark, so no MYR Sharpe ratio is reported (Federal Reserve Bank of St. Louis, n.d.).

Finding for RQ3.

> The investment relevance of de-dollarization lies primarily in changing the set of risks investors may wish to diversify, rather than proving that dollar assets will underperform. Historical evidence supports modest gold allocations as a potential means of improving risk-adjusted performance and drawdown resilience, but the effect depends materially on sample period, asset correlations and the source of funding. The strongest portfolio response is therefore diversification across defensive mechanisms, not wholesale substitution of gold for Treasury securities.

Broader global and institutional interpretation

De-dollarization is broader than reserve securities. BRICS governments have also explored payment mechanisms that can reduce the need for the dollar as an **intermediary currency** even where reserve portfolios remain dollar-heavy. That distinction matters: reducing dollar use in settlement is not the same economic event as selling a Treasury bond.

Project mBridge illustrates the difference. The BIS reports that the platform reached minimum-viable-product status in mid-2024, with partners from China, Hong Kong, Thailand and the UAE, joined by Saudi Arabia in 2024. Its aim was to test cross-border CBDC payments and foreign-exchange settlement. The BIS handed the project to its partners in October 2024. This is a payments-infrastructure milestone, not evidence of Treasury liquidation or a common BRICS reserve currency (Bank for International Settlements, 2024, n.d.).

The October 2024 finance-minister and central-bank-governor statement describes the BRICS Cross-Border Payments Initiative as voluntary and non-binding. The July 2025 Rio Declaration, paragraph 50, continues discussion of payment-system interoperability. These statements support an institutional diversification interpretation; they do not establish an operational common replacement currency (BRICS, 2024, section 5.1; BRICS, 2025, paragraph 50).

The New Development Bank provides a more tangible example. Its 2022–2026 strategy targets **30% of financing commitments over that five-year period in member-country currencies**. Its 2024 annual report states that RMB-, rand- and rupee-denominated financing represented **24.2% of its portfolio at end-2024**. The latter is a stock share, whereas the target concerns financing over a period: subtracting them to claim a 5.8-point performance shortfall would compare different denominators and time windows. Local-currency financing can reduce borrowers' currency mismatch, but does not imply matching Treasury sales by their central banks (New Development Bank, 2022, printed p. 17; New Development Bank, 2025, printed p. 49).

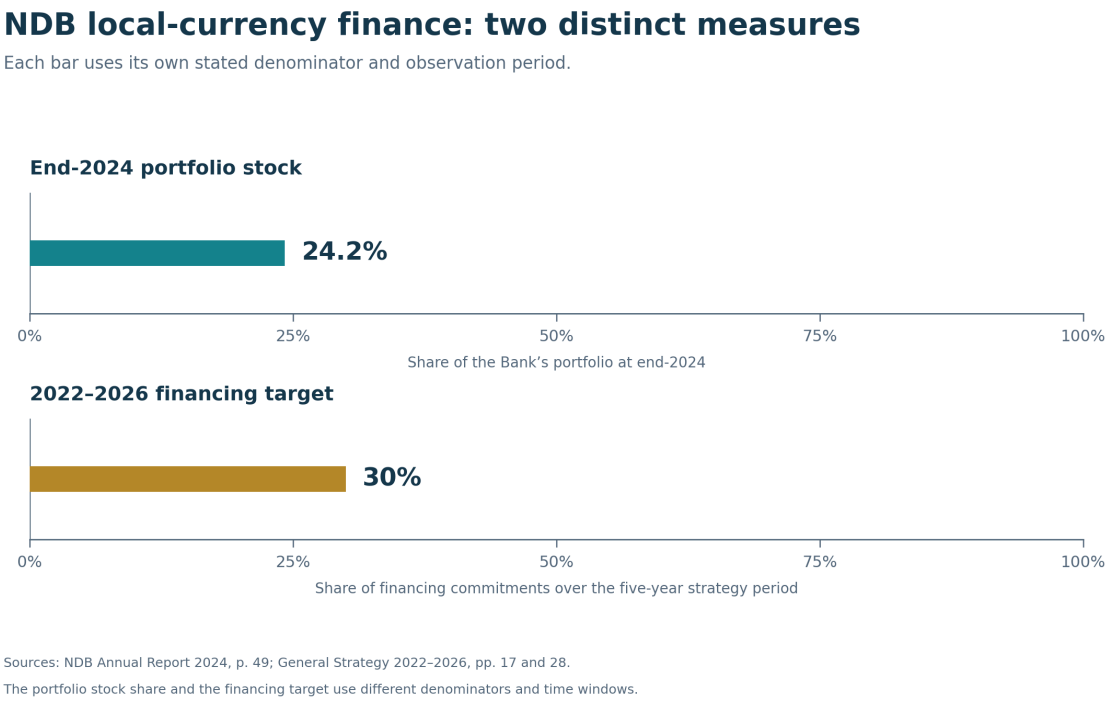


Figure 25. Author chart of the two published NDB percentages. Separate horizontal-bar panels with visible percentage axes identify the distinct denominators and periods; the chart is not a target-completion gauge (New Development Bank, 2022, 2025). Background gridlines are disabled.

The investment implication is that **payment de-dollarization, reserve diversification, and Treasury divestment are related but distinct variables**. A country may settle more bilateral trade in local currencies while continuing to hold Treasuries, buy gold without reducing dollar assets, or reduce Treasuries while retaining other dollar securities. The institutional evidence therefore complements the country data without establishing a coordinated reserve-sale mechanism.

Portfolio decision framework

The resulting framework connects observation to a decision without treating reserve headlines as a trading signal. First assess whether the evidence identifies a change in net Treasury demand relative to supply, then stress-test actual duration exposure. Separately evaluate whether a proposed gold allocation improves portfolio covariance, drawdown and liquidity under plausible regimes, including costs and the asset used to fund it. Uncertain demand evidence is a reason to monitor exposure, not evidence of zero repricing risk (Dilts Stedman, 2025; Doshi et al., 2026; Markowitz, 1952).

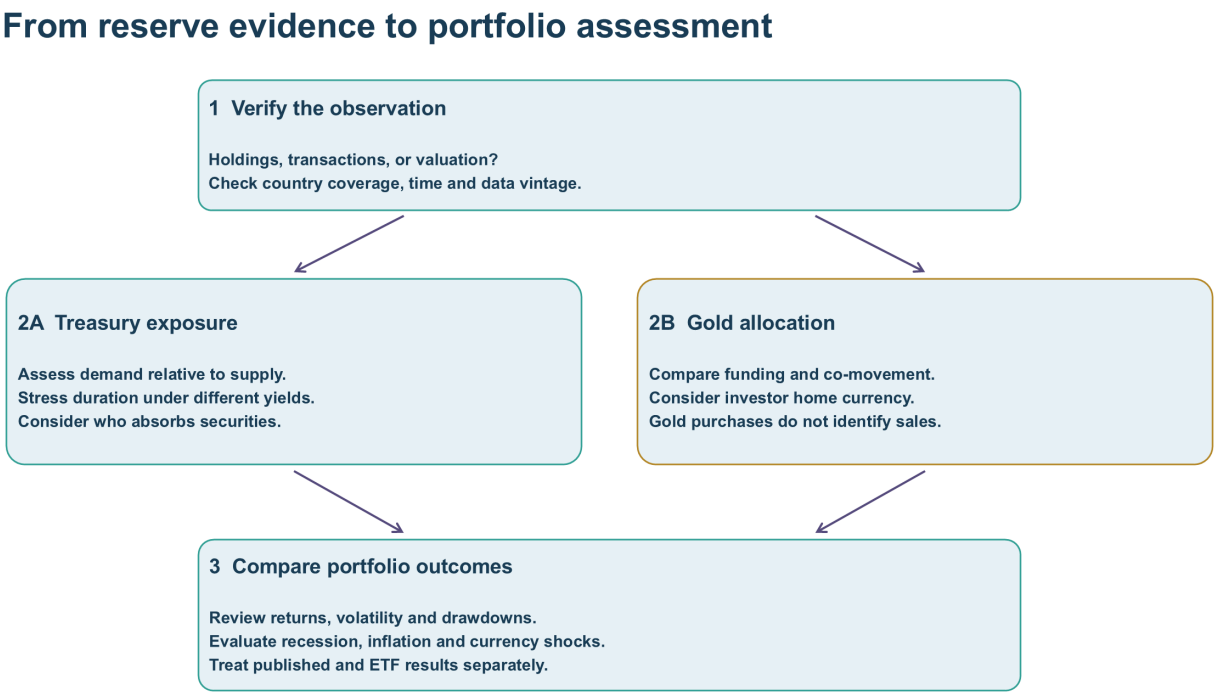


Figure 26. Author synthesis of the reviewed evidence, illustrating the portfolio assessment process. Arrows indicate an assessment sequence, not estimated causal coefficients or an investment recommendation. Both Treasury and gold decisions require portfolio-specific evaluation.

Evidence classification

To prevent the final report from blending evidence and conjecture, the key claims should be classified explicitly.

Claim	Classification	Strength
China's reported Treasury holdings fell from US$1.124tn in 2018 to US$684.4bn in 2025	**Observed fact**	High
Brazil's reported Treasury holdings fell materially	**Observed fact**	High
India's holdings ended 2025 above 2018	**Observed fact**	High
Dollar share of global FX reserves was 56.52% in 2025 Q4, September 2026 release	**Published revised aggregate**	High; vintage-specific
China cumulative 2018–2025 long-term net transactions were −US$533.557bn	**Published/estimated CSLT decomposition**	Substantial descriptive evidence; not identified sovereign trading
Treasury sales funded the reported gold accumulation	**Unsupported funding attribution**	Not established
Domestic physical gold is immune to sanctions and equivalent to GLD	**Unsupported generalization**	Reject
92% of 2025 Q2 dollar-share decline was due to FX valuation	**Observed IMF decomposition**	High

Claim	Classification	Strength
2025 central-bank gold purchases were about 863t	**Estimated observed aggregate**	Moderate/high
BRICS sales raised the 10-year Treasury yield by 57 bp	**Unsupported as stated**	Reject
A 57-bp yield shock would reduce a duration-7.5 bond by roughly 4.28%	**Modeled scenario**	High mathematically; conditional economically
Some Chinese holdings may appear through foreign custodians	**Plausible inference**	Moderate
Changes in foreign official demand can influence Treasury yields	**Empirically supported mechanism**	High
De-dollarization will cause the dollar to collapse	**Speculative inference**	Very weak
Gold will always outperform bonds in geopolitical crises	**Unsupported generalization**	Reject
A modest gold sleeve can improve some portfolios	**Conditional historical finding**	Moderate/high
2018–2025 ETF results show an improvement for the tested gold allocations	**Original historical calculation**	Sample- and implementation-specific; no causation or forecast
Stablecoin inflows affect three-month bill yields in the BIS study	**Published model estimate**	Model-, maturity- and sample-specific; not a BRICS coefficient
2018–2025 foreign-official long-term net purchases total US$78.9bn	**CSLT descriptive calculation**	Estimated and directly reported observations span a reporting change

Chronology, not a causal flow. The timeline separates observation years from publication milestones. Its year labels are schematic rather than proportionally spaced. In particular, the 2025 gold-demand and Q4 reserve-share observations were released in 2026; the earlier October 2025 IMF article concerns a different publication vintage. The 2023 entry identifies the TIC reporting change (Bank for International Settlements, 2024; International Monetary Fund, 2026b; U.S. Department of the Treasury, n.d.; World Gold Council, 2026a).



Figure 27. Reserve diversification chronology, showing observation and publication milestones. Each year has one list of observation or publication milestones. Sources and interpretation are as described above.

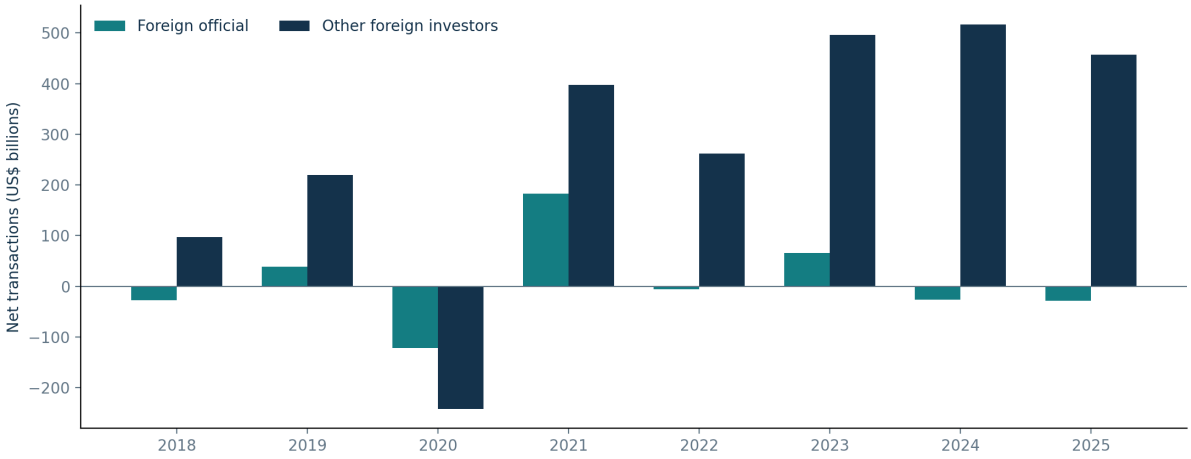
The vertical chronology orders observations and releases; it does not show that an earlier event caused a later one. The separate conceptual transmission diagram is also conditional: reserve reallocation toward gold may accompany lower Treasury demand, but it is not a necessary consequence of a change in reported holdings.

Supporting figures

The figures below accompany the findings above. Sources and calculation choices are identified in each caption and in the underlying data.

Foreign investors followed different transaction paths

Long-term U.S. Treasuries | Net purchases (+) and net sales (−) | 2018–2025

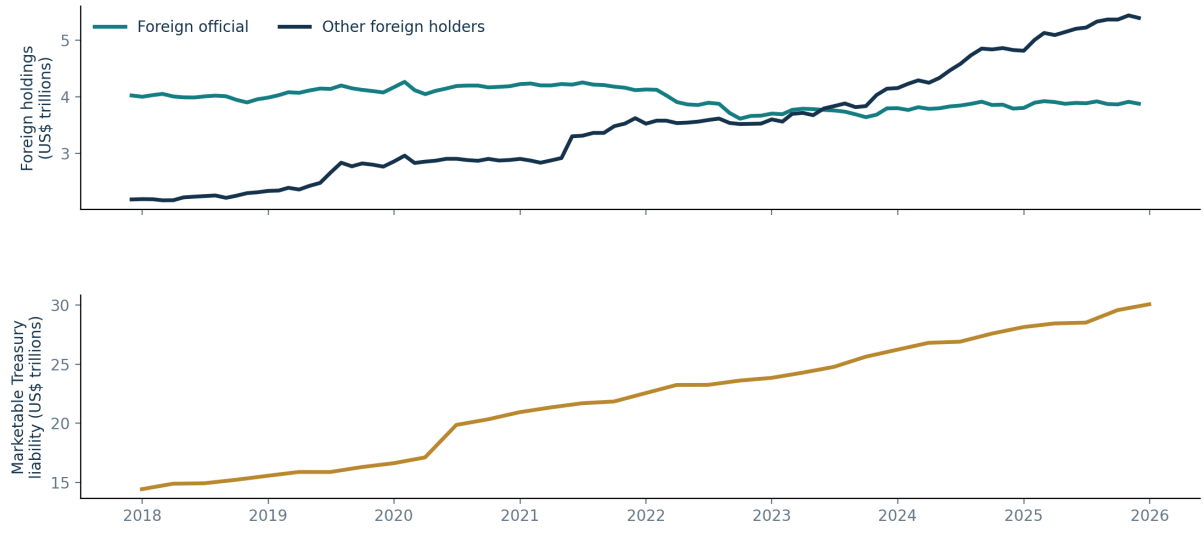


Source: U.S. Treasury TIC/CSLT via FRED. Before February 2023, transactions are estimates; later observations are directly reported. Other foreign investors exclude U.S. investors and are calculated as all foreign less foreign official.

Figure 28. Foreign-official and other foreign net transactions, 2018–2025. The latter is total foreign less official, not domestic private demand. CSLT transactions through January 2023 are estimated.

Treasury supply and foreign holdings grew at different rates

Separate measures | Year-end 2017 through 2025 | Levels, not a flow decomposition



Source: TIC/CSLT via FRED; Federal Reserve Financial Accounts FGTSL. The series differ in coverage and valuation. These stock levels are not subtracted to estimate domestic investors or issuance flows.

Figure 29. Foreign holdings and marketable Treasury liabilities have different measurement coverage; they are not a balance-sheet identity.

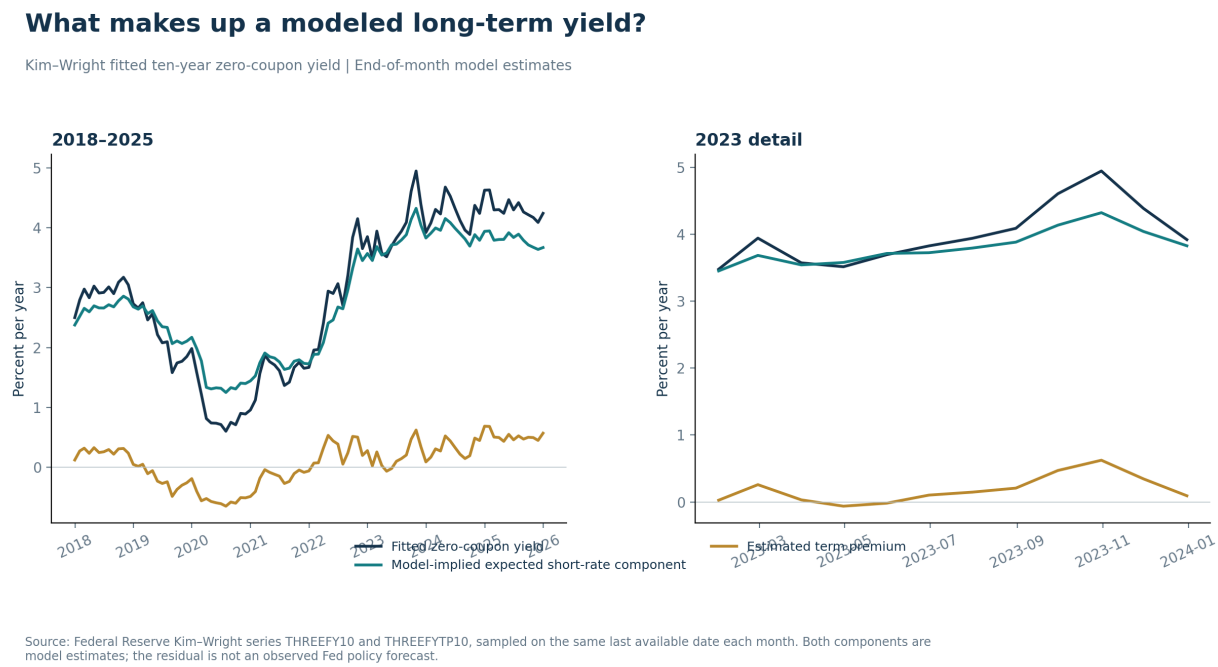


Figure 30. Kim–Wright model estimates sampled at month end; the residual is not an observed policy forecast.

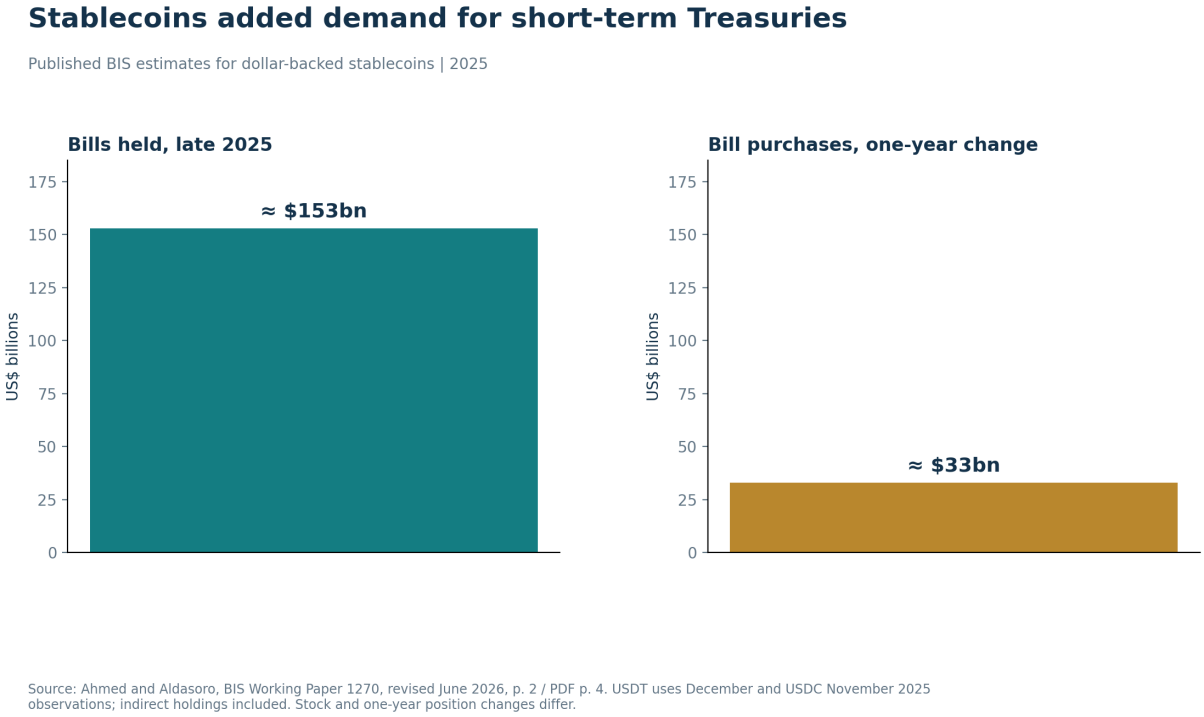


Figure 31. Year-end holdings and annual purchases are distinct measures. Based on Ahmed and Aldasoro (2026), June revision.

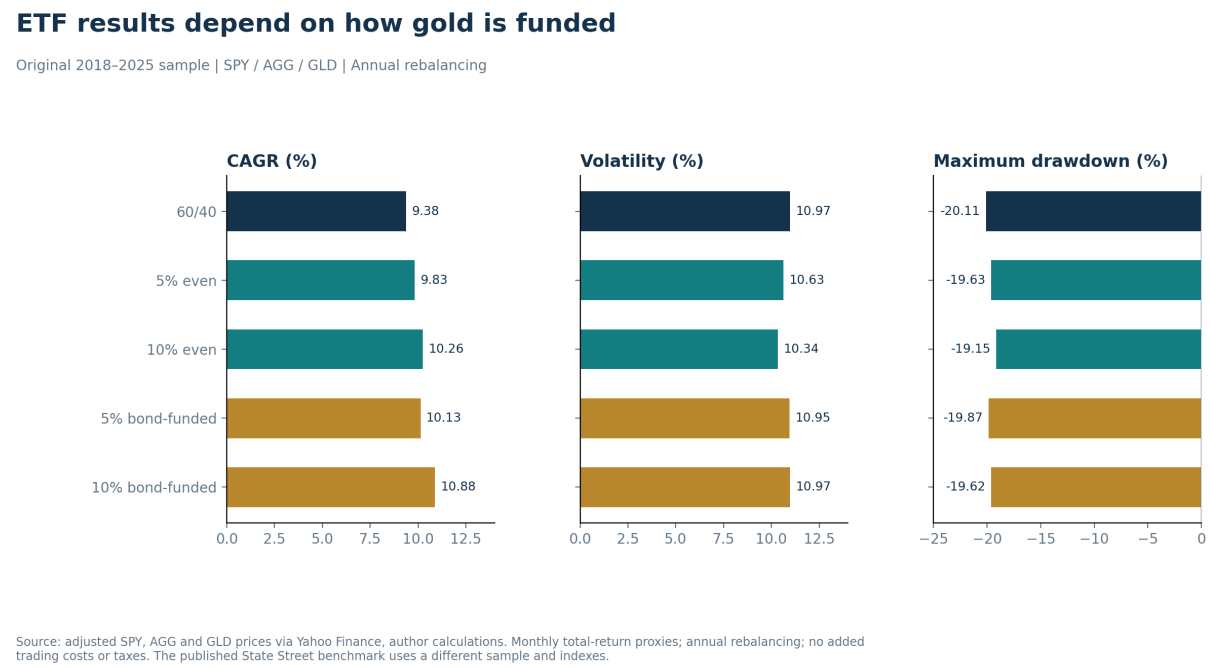


Figure 32. Monthly 2018–2025, USD, AGG bond proxy; BIL cash proxy. No additional transaction costs or taxes.

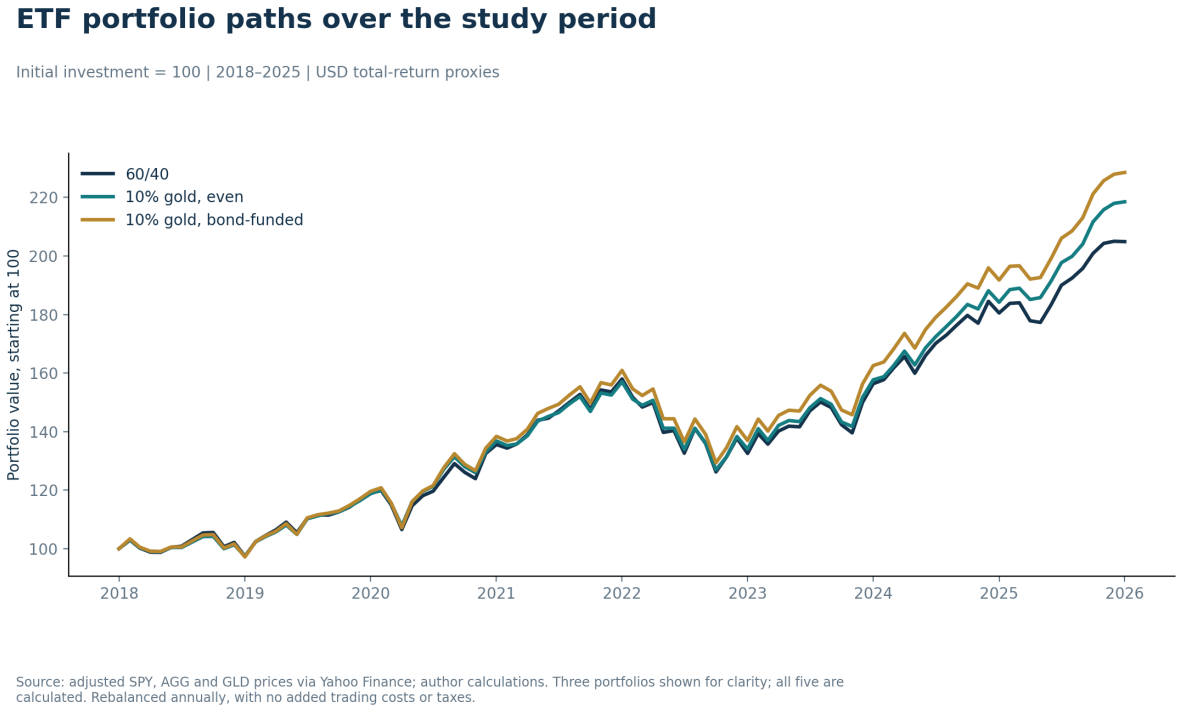


Figure 33. December 2017 starting value = 100; annual rebalancing. Monthly end values do not capture intramonth troughs.

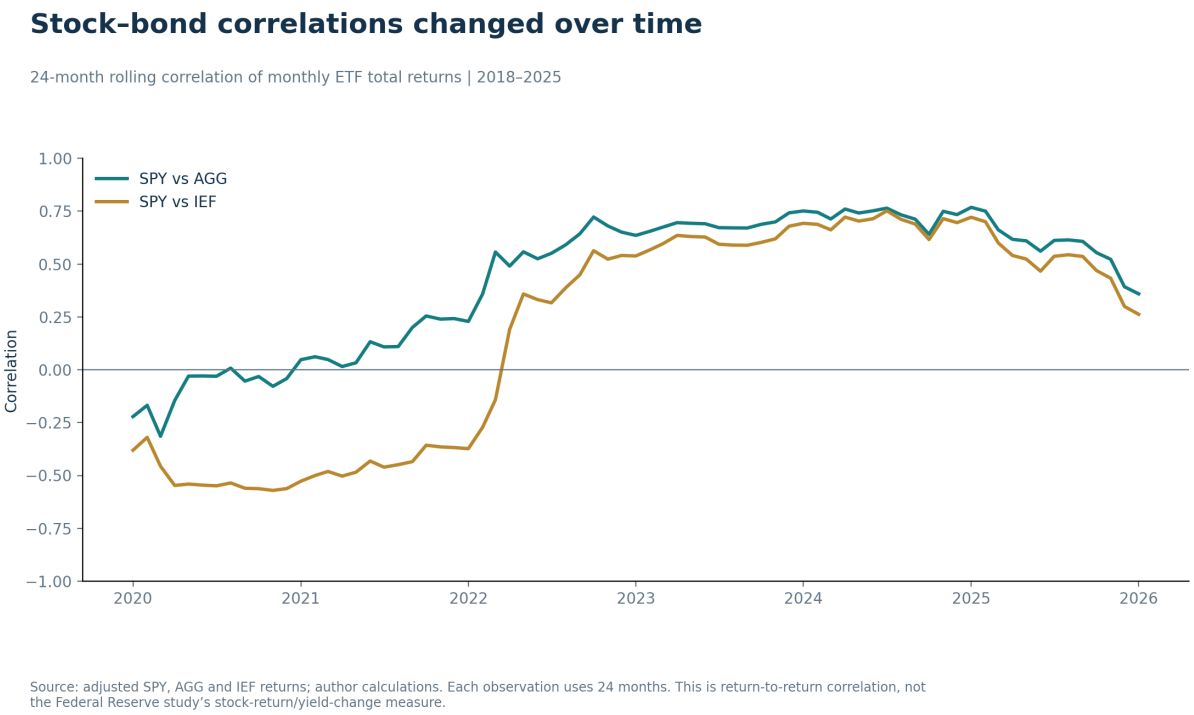


Figure 34. 24-month rolling correlations use monthly ETF returns. Results describe the sample and are not an identified causal effect.

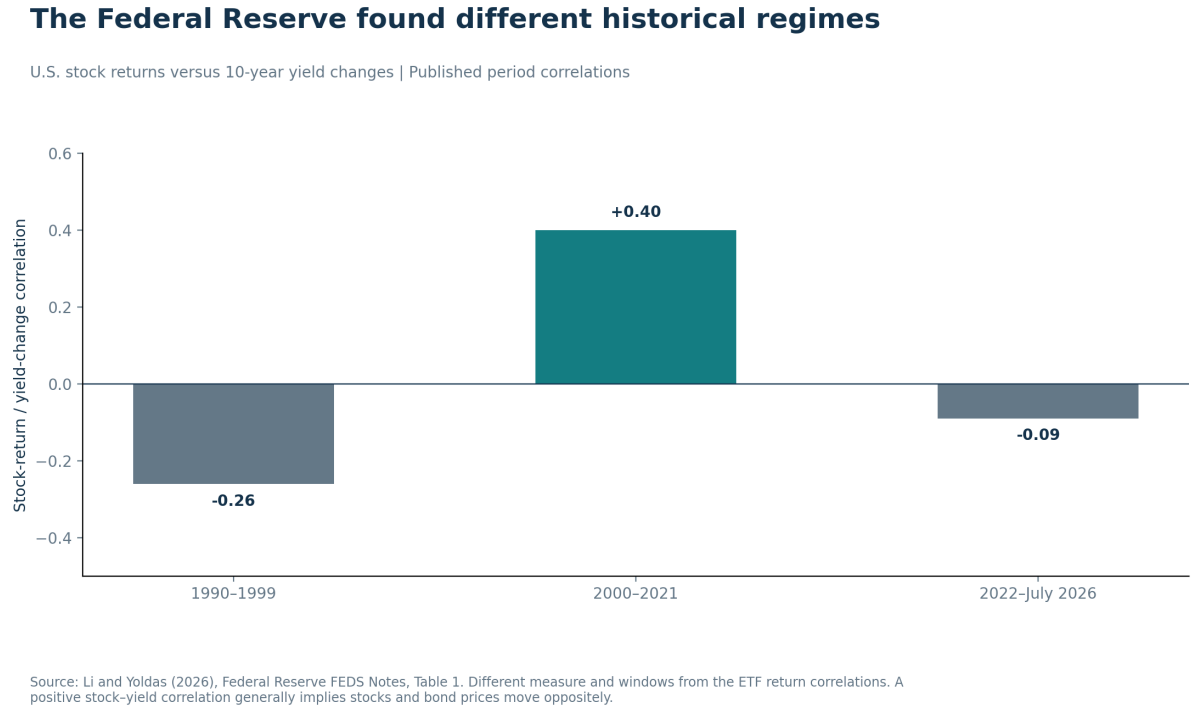


Figure 35. Li and Yoldas (2026), Table 1. This is not the same measure as return-to-return correlations; the final period extends into 2026.

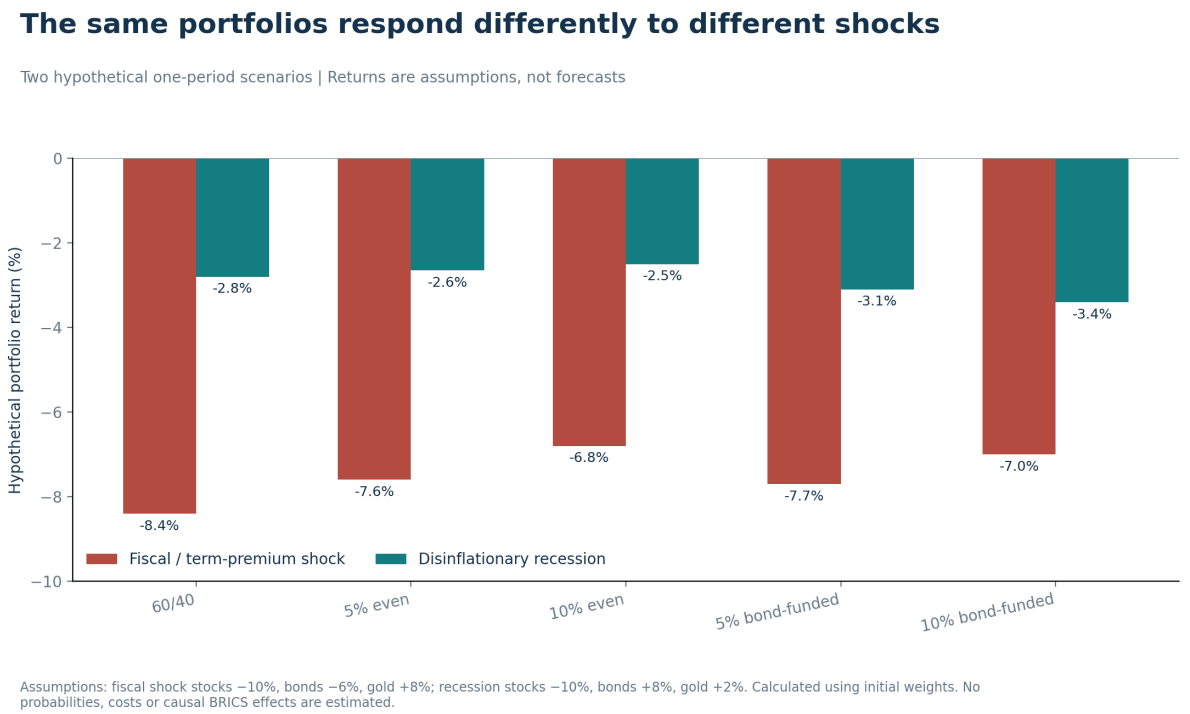


Figure 36. Hypothetical stock, bond and gold returns are stated in the figure. These are not forecasts or measured BRICS effects.

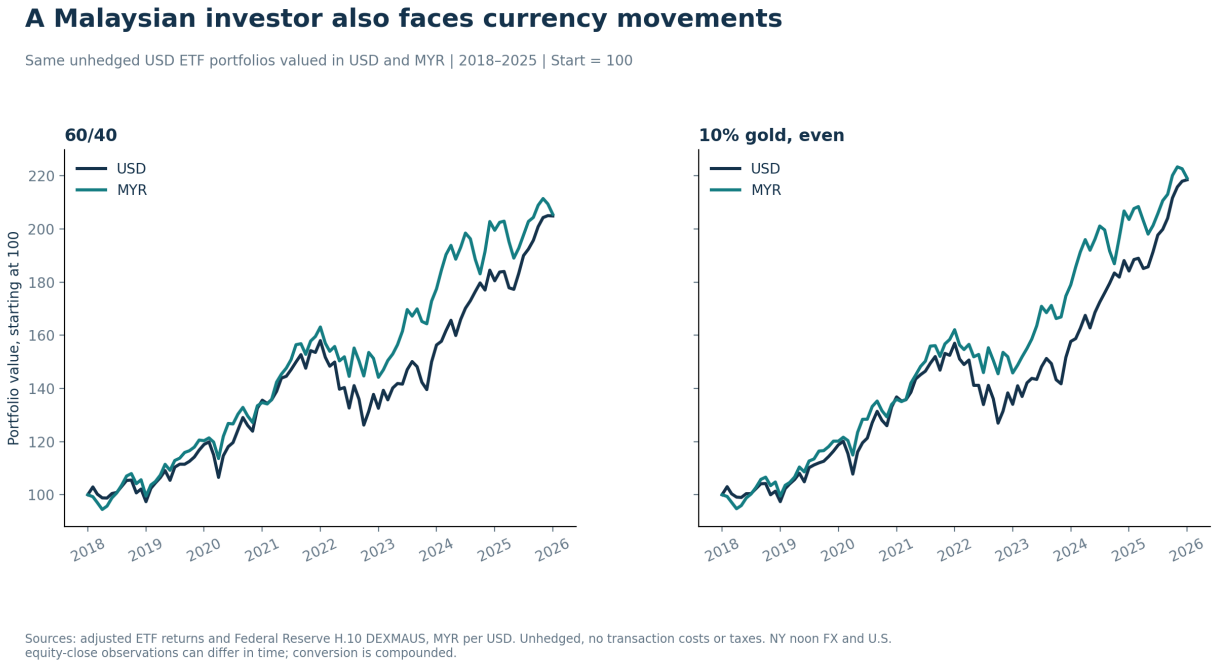


Figure 37. Same portfolios converted at MYR per USD exchange rates. The time of the FX observation differs from U.S. equity close.

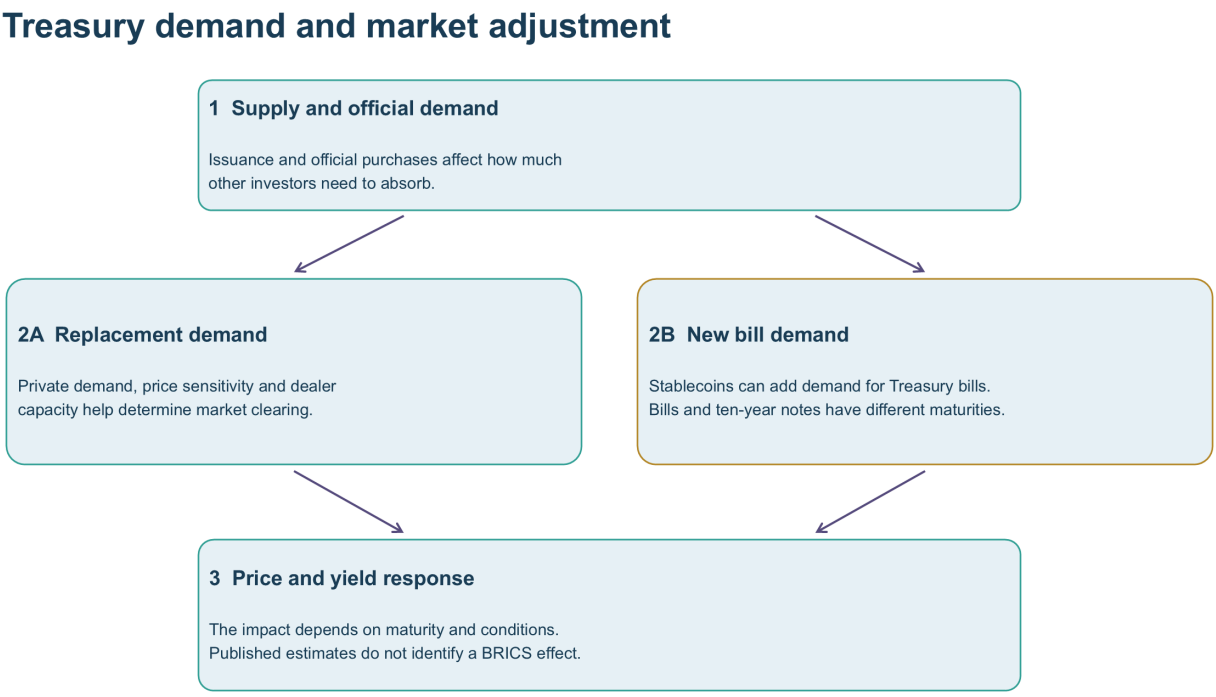


Figure 38. Native PowerPoint SmartArt with editable connectors. Arrows show a conceptual sequence, not measured causal coefficients.

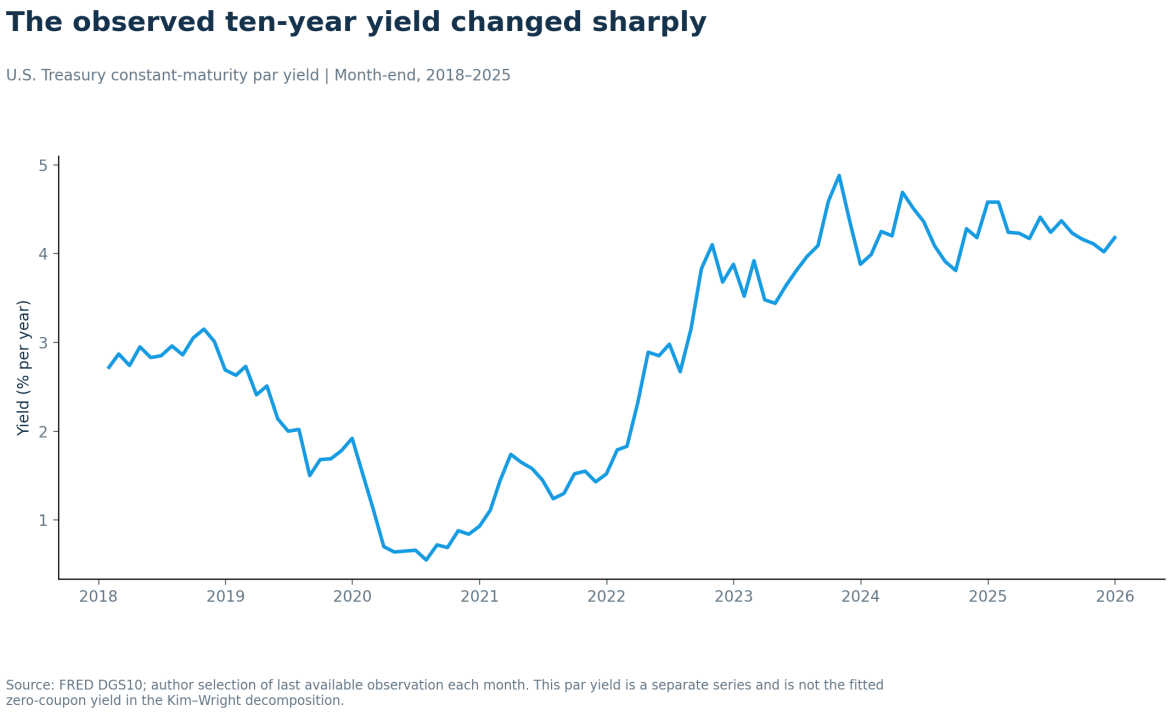


Figure 39. Month-end DGS10 par yield. This is a different series from the fitted zero-coupon model yield in Figure 30.

Conclusion and Reflection

The evidence supports the existence of a **de-dollarization movement**, but the term needs a narrower definition than is often used in commentary. Between 2018 and 2025, China and Brazil clearly reduced their directly reported U.S. Treasury holdings, Russia's reported holdings became very small, and central-bank gold demand accelerated globally. Yet India did not follow the same Treasury trajectory, total foreign holdings of U.S. Treasuries increased substantially, and the dollar remained the largest reserve currency by a wide margin at the end of 2025 (International Monetary Fund, 2026a; U.S. Department of the Treasury, 2026c; World Gold Council, 2026a).

The decomposition shows that net sales were the dominant component of China's long-term decline, while identifying 2022 as a year with particularly large valuation losses. India's opposite 2022 transaction direction and later reversal reinforce the heterogeneous-country conclusion. Gold quantity growth and domestic storage changes document additional dimensions of reserve management, but do not trace Treasury-sale proceeds or justify equating sovereign bullion with an ETF. The revised global reserve data likewise show a smaller dollar share without a comparable gain in the yuan's share, and without a fall in the dollar value of reported global dollar reserves (International Monetary Fund, 2026a; McCallum et al., 2026).

The first research objective is therefore achieved only with an important qualification. **Reserve diversification is real; a synchronized BRICS liquidation is not demonstrated.** This distinction matters because an investment report should not transform a geopolitical narrative into a financial fact merely because both point in the same direction.

The second research objective produces an equally conditional result. Foreign official demand has a measurable role in Treasury pricing, but country holdings cannot isolate the causal BRICS contribution to the 10-year yield. The empirical literature suggests that large foreign flow shocks can produce economically meaningful yield changes, while the duration analysis demonstrates how such a change would affect existing bond investors. Yet the same period includes major changes in Federal Reserve policy, inflation, fiscal issuance and term premiums. A precise statement that "BRICS selling caused X basis points of the increase in yields" would therefore exceed the identification capacity of the available evidence (Dilts Stedman, 2025).

The third research objective is best answered through portfolio construction rather than political prediction. Modern Portfolio Theory does not require investors to decide whether the dollar or BRICS will "win." It requires them to examine expected return, volatility and covariance. Gold is relevant because its return drivers differ from those of nominal bonds and equities, and historical evidence indicates that modest allocations have sometimes improved portfolio Sharpe ratios and drawdowns. But gold does not generate contractual coupon income, can be volatile and is not a reliable safe haven in every period. Markowitz's central insight remains more useful than slogans about reserve currencies: portfolio value depends on how assets interact, not on how impressive each asset sounds in isolation (Markowitz, 1952, pp. 77–91) (Baur & Lucey, 2010; Markowitz, 1952).

The most defensible investment interpretation is therefore **risk diversification rather than dollar abandonment**. For investors exposed heavily to long-duration dollar bonds, gradual weakening in structural foreign-official demand is an additional factor to monitor because it can raise the term premium

required to clear the Treasury market. A modest gold allocation may hedge some inflationary, geopolitical and monetary-regime risks. At the same time, retaining high-quality U.S. fixed income preserves liquidity, contractual income and recession-related diversification that gold does not replicate.

The additional calculations deepen both conclusions. Treasury-market adjustment depends on replacement demand, maturity and the economic state, while the original ETF exercise shows that funding and investor currency change portfolio outcomes. The 2018–2025 ETF results and the longer published State Street comparison are different historical experiments, not evidence of a universal optimal gold weight.

De-dollarization is gradual and differs between countries. Its effect on Treasuries depends on the circumstances. Its portfolio implication is diversification.

References

Ahmed, R., & Aldasoro, I. (2026). *Stablecoins and safe asset prices* (BIS Working Paper No. 1270, revised June 2026). Bank for International Settlements.

Ahmed, R., & Rebucci, A. (2024). Dollar reserves and U.S. yields: Identifying the price impact of official flows. *Journal of International Economics, 152*, Article 103974. https://doi.org/10.1016/j.jinteco.2024.103974

Arslanalp, S., Eichengreen, B., & Simpson-Bell, C. (2022). *The stealth erosion of dollar dominance: Active diversifiers and the rise of nontraditional reserve currencies* (IMF Working Paper No. 22/58). International Monetary Fund.

Bank for International Settlements. (2024, June 5). *Project mBridge reaches minimum viable product stage and invites further international participation*.

Bank for International Settlements. (n.d.). *Project mBridge*. Retrieved November 4, 2026.

Bank of Russia. (2019). *Bank of Russia annual report for 2018*. Chart 32, printed/PDF p. 86.

Bank of Russia. (2020). *Bank of Russia annual report for 2019*. Chart 37, printed p. 99 (PDF p. 100).

Bank of Russia. (2022). *Bank of Russia annual report for 2021*. Chart 32, printed p. 102 (PDF p. 104).

Baur, D. G., & Lucey, B. M. (2010). Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold. *The Financial Review, 45*(2), 217–229. https://doi.org/10.1111/j.1540-6288.2010.00244.x

Beltran, D., & Li, C. (2026). *Estimating yield impacts of Treasury demand and supply changes* (International Finance Discussion Paper No. 1447). Board of Governors of the Federal Reserve System. https://doi.org/10.17016/IFDP.2026.1447

Bertaut, C., von Beschwitz, B., & Curcuru, S. (2025, July 18). *The international role of the U.S. dollar—2025 edition*. FEDS Notes. Board of Governors of the Federal Reserve System.

Board of Governors of the Federal Reserve System. (n.d.). *Three-factor nominal term structure model*. Retrieved November 4, 2026.

BRICS. (2024, October 11). *BRICS finance ministers and central bank governors joint statement*. Section 5.1.

BRICS. (2025, July 6). *Rio de Janeiro declaration*. Paragraph 50.

cjseven007. (2026). *FOI analysis: De-dollarisation* [Source code and archived data, commit e8cd1fab26a2a810b8dd72fb85d55d28cd0906c3]. GitHub.

Diercks, A. M., & Asnani, D. (2024, September 3). *The Treasury tantrum of 2023*. FEDS Notes. Board of Governors of the Federal Reserve System.

Dilts Stedman, K. (2025, July 9). *The changing investor composition of U.S. Treasuries, Part 1: Foreign Treasury sales could raise U.S. yields*. *Federal Reserve Bank of Kansas City Economic Bulletin*, 1–4.

Doshi, A., Abukhalaf, M., & Andrade, D. (2026, October 21). *Rethinking the 60/40 portfolio: The case for gold*. State Street Investment Management. Monthly historical simulation, December 31, 2004–December 31, 2025.

Federal Reserve Bank of New York. (n.d.). *Treasury term premia*. Retrieved November 4, 2026.

Federal Reserve Bank of St. Louis. (n.d.). *FRED: DEXMAUS, FGTSL, THREEFY10 and THREEFYTP10* [Data sets]. Retrieved November 6, 2026.

International Monetary Fund. (2026a, September 30). *Currency composition of official foreign exchange reserves: September 30, 2026 release* [Data set]. Quarterly observations used: 2018 Q4–2025 Q4. Legacy indicator label "Allocated foreign exchange reserves"; revised world total includes imputed allocations. Official methodology is listed separately.

International Monetary Fund. (2026b, March 27). *IMF data brief: Currency composition of official foreign exchange reserves—Fourth quarter 2025*.

International Monetary Fund. (n.d.). *Currency composition of official foreign exchange reserves (COFER)* [Data set and methodology]. Retrieved November 4, 2026.

Kwende, G., Nephew, E., & Sánchez-Muñoz, C. (2025, October 1). *Dollar's share of reserves held steady in second quarter when adjusted for FX moves*. International Monetary Fund.

Li, C., & Yoldas, E. (2026, October 16). *Government bonds' safe-haven property and inflationary pressures*. FEDS Notes. Board of Governors of the Federal Reserve System.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance, 7*(1), 77–91. https://doi.org/10.1111/j.1540-6261.1952.tb01525.x

McCallum, A. H., DeMane, L., Garcia, A., Gillette, J., Grover, T., Haw, H., Hinshaw, H., Kim, N., & Loucky, A. (2026, May 21). *The CSLT: Unifying U.S. cross-border securities holdings and transactions data*. FEDS Notes. Board of Governors of the Federal Reserve System. https://doi.org/10.17016/2380-7172.4084

New Development Bank. (2022). *General strategy for 2022–2026: Scaling up development finance for a sustainable future*. Printed pp. 17, 28 (PDF pp. 19, 30).

New Development Bank. (2025). *Annual report 2024*. Printed p. 49.

Office of Foreign Assets Control. (2022, June 28). *Frequently asked question 1029: Gold-related transactions and Russia-related sanctions*. U.S. Department of the Treasury. Originally issued March 24, 2022; updated June 28, 2022.

Reserve Bank of India. (2019, May 21). *Half yearly report on management of foreign exchange reserves: October–March 2018–19*. Section I.6, printed p. 7 (PDF p. 8).

Reserve Bank of India. (2025, October 28). *Half yearly report on management of foreign exchange reserves: April–September 2025*. Section I.6, printed p. 7 (PDF p. 8).

U.S. Department of the Treasury. (2026a). *Continuous Securities Long-Term (CSLT): Treasury positions, transactions and valuation changes* [Data set]. Treasury International Capital System, with Federal Reserve estimates; distributed through FRED. Retrieved November 2026.

U.S. Department of the Treasury. (2026b). *Major foreign holders of Treasury securities: Current Table 5* [Data set]. Treasury International Capital System. Retrieved November 4, 2026; used only for the stated vintage comparison.

U.S. Department of the Treasury. (2026c). *Major foreign holders of Treasury securities: Historical tables* [Data set]. Treasury International Capital System. Historical December observations, 2018–2025; retrieved November 4, 2026.

U.S. Department of the Treasury. (n.d.). *Frequently asked questions regarding the TIC system: Page 2*. Retrieved November 4, 2026.

Warnock, F. E., & Warnock, V. C. (2009). International capital flows and U.S. interest rates. *Journal of International Money and Finance, 28*(6), 903–919. https://doi.org/10.1016/j.jimonfin.2009.03.002

Weiss, C. (2026, September 3). *Why gold didn't actually overtake Treasury securities as the world's favorite reserve asset*. FEDS Notes. Board of Governors of the Federal Reserve System.

Wolcott, E. L. (2020). Impact of foreign official purchases of US Treasuries on the yield curve. *AEA Papers and Proceedings, 110*, 535–540. https://doi.org/10.1257/pandp.20201124

World Gold Council. (2019, January 31). *Gold demand trends: Full year and Q4 2018—Central banks and other institutions*. Contemporaneous publication vintage.

World Gold Council. (2025, June 17). *Central bank gold reserves survey 2025*.

World Gold Council. (2026a, January 29). *Gold demand trends: Q4 and full year 2025—Central banks*. Full publisher report pp. 16–17.

World Gold Council. (2026b, January 8). *Gold market commentary: Precious metal thunder*. Table 1: end-2025 LBMA Gold Price PM, US$4,368 per troy ounce.

World Gold Council. (n.d.). *Central banks gold reserves by country* [Data set and documentation]. Retrieved November 4, 2026. Quarterly workbook observations used: 2018 Q4–2025 Q4; the workbook export date is not established.

Yahoo Finance. (n.d.). *Historical adjusted ETF prices: SPY, AGG, IEF, GLD and BIL* [Data sets]. Retrieved November 2026; returns are calculated from preserved adjusted prices.